

Thank you for purchasing this Esco Biological Safety Cabinet. Please read this manual thoroughly to familiarize yourself with the many unique features and exciting innovations we have built into your new equipment. Esco provides many other resources at our website, www.escoglobal.com, to complement this manual and help you enjoy many years of productive and safe use of your Esco products.



Service Manual

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Labculture®

Class II Type A2

Biological Safety Cabinet

Copyright Information

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"Material in this manual is provided for informational purposes only. The contents and the product described in this manual (including any appendix, addendum, attachment or inclusion), are subject to change without notice. Esco makes no representations or warranties as to the accuracy of the information contained in this manual. In no event shall Esco be held liable for any damages, direct or consequential, arising out of or related to the use of this manual."

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Warranty Terms and Conditions

Esco products come with a limited warranty. The warranty period will vary depending on the product purchased, beginning on the date of shipment from any Esco international warehousing location. To determine which warranty applies to your product, refer to the appendix below.

Esco's limited warranty covers defects in materials and workmanship. Esco's liability under this limited warranty shall be, at our option, to repair or replace any defective parts of the equipment, provided that these parts, if proven to the satisfaction of Esco, were defective at the time of being sold and that all defective parts shall be returned, properly identified with a Return Authorization.

This limited warranty covers parts only, and not transportation/insurance charges.

This limited warranty does not cover:

- Freight or installation (inside delivery handling) damage. If your product was damaged in transit, you must file a claim directly with the freight carrier.
- Products with missing or defaced serial numbers.
- Products for which Esco has not received payment.
- Problems that result from:
 - External causes such as accident, abuse, misuse, problems with electrical power, improper operating environmental conditions.
 - Servicing not authorized by Esco.
 - Usage that is not in accordance with product instructions.
 - Failure to follow the product instructions.
 - Failure to perform preventive maintenance.
 - Using accessories, parts, or components not supplied by Esco.
 - Damage by fire, floods, or acts of God.
 - Customer modifications to the product.
- Consumables such as filters (HEPA, ULPA, carbon, pre-filters) and fluorescent / UV bulbs.
- Esco is not liable for any damage incurred on the objects used on or stored in Esco equipment. Users are advised to conduct risk assessment and add safety protocols based on their application and sample.

Factory installed, customer specified equipment or accessories are warranted only to the extent guaranteed by the original manufacturer. The customer agrees that in relation to these products purchased through Esco, our limited warranty shall not apply and the original manufacturer's warranty shall be the sole warranty in respect of these products. The customer shall utilize that warranty for the support of such products and in any event not look to Esco for such warranty support.

Esco encourages all users to register their equipment online at www.escoglobal.com/warranty_registrations.php or complete the warranty registration form included with each product.

ALL EXPRESS AND IMPLIED WARRANTIES FOR THE PRODUCT, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES AND CONDITIONS OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE LIMITED IN TIME TO THE TERM OF THIS LIMITED WARRANTY. NO WARRANTIES, WHETHER EXPRESS OR IMPLIED, WILL APPLY AFTER THE LIMITED WARRANTY PERIOD HAS EXPIRED. ESCO DOES NOT ACCEPT LIABILITY BEYOND THE REMEDIES PROVIDED FOR IN THIS LIMITED WARRANTY OR FOR SPECIAL, INDIRECT, CONSEQUENTIAL OR INCIDENTAL DAMAGES, INCLUDING, WITHOUT LIMITATION, ANY LIABILITY FOR THIRD-PARTY CLAIMS AGAINST YOU FOR DAMAGES, FOR PRODUCTS NOT BEING AVAILABLE FOR USE, OR FOR LOST WORK. ESCO'S LIABILITY WILL BE NO MORE THAN THE AMOUNT YOU PAID FOR THE PRODUCT THAT IS THE SUBJECT OF A CLAIM. THIS IS THE MAXIMUM AMOUNT FOR WHICH ESCO IS RESPONSIBLE.

These Terms and Conditions shall be governed by and construed in accordance with the laws of Singapore and shall be subject to the exclusive jurisdiction of the courts of Singapore.

Technical Support, Warranty Service Contacts

USA: 1 215-441-9661

Singapore: +65 6542 0833

Global Email Helpdesk: support@escoglobal.com

Visit <http://escoglobal.com> to talk to a Live Support Representative

Distributors are encouraged to visit the Distributor Intranet for self-help materials.

Product Appendix, Warranty Listings

Biological Safety Cabinets, Laminar Flow Cabinets, Laboratory Animal Research Workstations, HEPA-Filtered Cabinets (except Streamline brand)	3 years limited
Laboratory Fume Hoods	2 years limited
Ductless Fume Hoods	2 years limited
Cleanroom Equipment	1 year limited
Laboratory Ovens and Incubators	1 year limited
Laboratory Shaker	1 year limited
CO ₂ Incubators	2 years limited
Containment/Pharma Products	2 years limited
Ultra-low Temperature Freezer	5 years limited 5 years on compressor
Laboratory Centrifuge	2 years limited
Freeze Dryer	2 years limited
Laboratory Refrigerator	2 years limited
Thermal Cyclers	3 years limited for MiniPro, Aeris, Provocell 2 years limited for Spectrum 2 years on blocks and Peltier units 2 years on all electronic components 3 years on the housing and fans

Note: The warranty periods may vary by country. Contact your local distributor for specific warranty details.

For international distributors, warranty period starts two months from the date the equipment is shipped from Esco facility. This allows shipping time so the warranty will go into effect at approximately the same time the equipment is delivered to the user. The warranty protection extends to any subsequent owner during the warranty period. Distributors who stock Esco equipment are allowed an additional four months for delivery and installation, providing the product is registered with Esco. User can register their products online at www.escoglobal.com/warranty or complete the warranty registration form include with each product.

Policy updated on 1st January 2015 (This limited warranty policy applies to products purchased on and after 1st January 2015)

Introduction

1. Products Covered

Esco Class II Type A2 Biological Safety Cabinet						
Model	Electrical Rating	0.9 meters (3 feet)	1.2 meters (4 feet)	1.5 meters (5 feet)	1.8 meters (6 feet)	2.4 meters (8 feet)
Labculture	220-240 V AC, 50Hz, 1Φ	LA2-3A1-E	LA2-4A1-E	LA2-5A1-E	LA2-6A1-E	LA2-8A1-E
		LA2-3A1-E-M	LA2-4A1-E-M	LA2-5A1-E-M	LA2-6A1-E-M	LA2-8A1-E-M
		LA2-3L1-E	LA2-4L1-E	LA2-5L1-E	LA2-6L1-E	LA2-8L1-E
		LA2-3L1-E-M	LA2-4L1-E-M	LA2-5L1-E-M	LA2-6L1-E-M	LA2-8L1-E-M
	110-125 V AC, 60Hz, 1Φ	LA2-3A2-E	LA2-4A2-E	LA2-5A2-E	LA2-6A2-E	LA2-8A2-E
		LA2-3A2-E-M	LA2-4A2-E-M	LA2-5A2-E-M	LA2-6A2-E-M	LA2-8A2-E-M
		LA2-3L2-E	LA2-4L2-E	LA2-5L2-E	LA2-6L2-E	LA2-8L2-E
		LA2-3L2-E-M	LA2-4L2-E-M	LA2-5L2-E-M	LA2-6L2-E-M	LA2-8L2-E-M
	220-240 V AC, 60Hz, 1Φ	LA2-3A3-E	LA2-4A3-E	LA2-5A3-E	LA2-6A3-E	LA2-8A3-E
		LA2-3A3-E-M	LA2-4A3-E-M	LA2-5A3-E-M	LA2-6A3-E-M	LA2-8A3-E-M
		LA2-3L3-E	LA2-4L3-E	LA2-5L3-E	LA2-6L3-E	LA2-8L3-E
		LA2-3L3-E-M	LA2-4L3-E-M	LA2-5L3-E-M	LA2-6L3-E-M	LA2-8L3-E-M

2. Safety Warning

- Anyone working with, on or around this equipment should read this manual. Failure to read, understand and comply with the instructions given in this manual may result in damage to the unit, injury to operating personnel, and / or poor equipment performance.
- Any internal adjustment, modification or maintenance to this equipment must be undertaken by qualified service personnel.
- The use of any hazardous materials in this equipment must be monitored by an industrial hygienist, safety officer or some other suitably qualified individuals.
- Explosive or inflammable substances should never be used in the cabinet unless adequate risk assessment has been carried out.
- If chemical, radiological or other non-microbiological hazards are being used in the cabinet, additional protective measures should be taken based on an adequate risk assessment.
- This cabinet should not be used with cytotoxic substances unless it has been determined that the filter can be safely changed. Please note that cytotoxic substances cannot be inactivated by conventional gaseous decontamination method (e.g. formaldehyde) used to inactivate biological agents.
- The biological hazard symbol on the front panel of the cabinet indicates the presence of biological substances that pose a threat to human health.
- Before you process, you should thoroughly understand the installation procedures and take note of the environmental / electrical requirements.
- In this manual, important safety related points will be marked with the symbol. 
- If the equipment is used in a manner not specified by this manual, the protection provided by this equipment may be impaired.

3. Document Management

We recommend that you keep this manual, along with the factory test report close to the cabinet for easy reference by the cabinet operator and qualified maintenance personnel.

If you require replacements for any of the provided documentation (including factory test reports) you can request copies from Esco Customer Services*. Please provide the following information when making requests for replacement documents:

- Company (Organization) Name
- Product Brand and Model
- Product Serial Number
- Documents requested

* There may be a nominal charge for this service.

4. Limitation of Liability

The disposal and / or emission of substances used in connection with this equipment may be governed by various local regulations. Familiarization and compliance with any such regulations are the sole responsibility of the users. Esco's liability is limited with respect to user compliance with such regulations.

5. European Union Directive on WEEE and RoHS

The European Union has issued two directives:

- **Directive 2002/96/EC on Waste Electrical and Electronic Equipment (WEEE)**

This product is required to comply with the European Union's Waste Electrical & Electronic Equipment (WEEE) Directive 2002/96/EC. It is marked with the following symbol:

Esco sells products through distributors throughout Europe. Contact your local Esco distributor for recycling/disposal.



Recommended method of disposal is according to The Federal, State and Local Government regulations.

- **Directive 2002/95/EC on Restriction on the use of Hazardous Substances (RoHS)**

With respect to the directive on RoHS, please note that this cabinet falls under category 8 (medical devices) and category 9 (monitoring and control instruments) and is therefore exempted from requirement to comply with the provisions of this directive.

6. Symbols

Information in this manual may be prefaced with the following symbols. They are provided to help you identify important operational, safety, maintenance or conformance issues.



Electrical Hazard: Danger of electric shock



Turn Off and Disconnect From Main Supply Before Proceeding: Do not perform this operation while the unit is operational



The Biohazard Symbol on the front panel of the cabinet indicates the presence of biological substances that pose a threat to human health

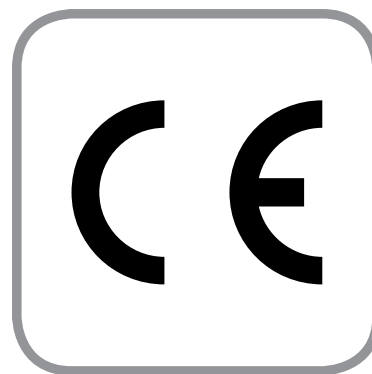


Approved Service Engineer Only: Operation to be performed only by approved engineers

Declaration of Conformity

In accordance to EN ISO/IEC 17050-1:2010

We, Esco Micro Pte.Ltd.
of 21 Changi South Street 1
Singapore, 486777
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declare on our sole responsibility that the product:

Category : Class II Type A2 Biological Safety Cabinet
Brand : Labculture
Model : LA2-3A1-E, LA2-4A1-E, LA2-5A1-E, LA2-6A1-E, LA2-8A1-E, LA2-3A1-E-M, LA2-4A1-E-M, LA2-5A1-E-M, LA2-6A1-E-M, LA2-8A1-E-M, LA2-3L1-E, LA2-4L1-E, LA2-5L1-E, LA2-6L1-E, LA2-3L1-E-M, LA2-4L1-E-M, LA2-5L1-E-M, LA2-6L1-E-M

in accordance with the following directives:

2014/35/EU : The Low Voltage Directive and its amending directives
2014/30/EU : The Electromagnetic Compatibility Directive and its amending directives
2011/65/EU : The RoHS in Electrical and Electronic Equipment Directive and its amending directives

has been certified independently to comply with the requirement of the following Harmonized Standards:

Low Voltage : EN 61010-1:2001
EMC : EN 61326-1:2006 Class B
Design/Performance Criteria : EN 12469 (2000) Class II Microbiological Safety Cabinet

More information may be obtained from Esco's authorized distributors located within the European Union. A list of these parties and their contact information is available on request from Esco.

Lim Lay Yew

Lim Lay Yew
Director, Esco

Chapter 1 - Maintenance and Re-certification of the Cabinet

Esco products generally provide years of trouble-free operation however like all equipment they require maintenance and service. Maintenance and service should be carried out by trained personnel. Esco offers training courses to equip service providers with the latest skills, information and tools to successfully maintain and service Esco products. For more information on the nearest training course, please contact Esco.

Service providers should familiarize themselves with the basic operating principles of products before working on them. Suitable references include information in this Manual or brochures which may be downloaded from the Esco web site.

Biological safety cabinets generally require:

- Re-certification, when:
 - The cabinet is re-located.
 - Cabinet performance is suspect.
 - After filter or blower replacement.
 - At least once a year or based on your risk assessment.
 - During recertification:
 - Cabinet airflow velocities and flow patterns are verified against the manufacturer's specifications and relevant international standards
 - Filters are scan-tested to ensure they do not leak.
 - Operator comfort tests may be performed.
 - If airflow velocities are found to be off setpoint, adjustments are made as part of the certification process before final values are recorded.
- Airflow alarm calibration, when:
 - The cause of the airflow alarm cannot be determined.
 - Re-certification indicates the displayed airflow deviates by > 0.02 m/s (4 fpm) from actual measured velocities (only for products with airflow velocity displays).
- Decontamination before filter or fan replacement, or as specified.
- Filter replacement, when:
 - The filters are clogged and the fan(s) are unable to compensate for the filter loading.
 - Filter leaks which cannot be repaired are found during scan-testing.

Note: Before filter replacement, the cabinet must be decontaminated.
- Fan replacement (typically rare) if failure occurs.
- Routine maintenance also includes:
 - Fluorescent lamp(s) replacement - typically once every 2 years.
 - UV lamp replacement - typically once every 1 year.

1.1. Installation Check

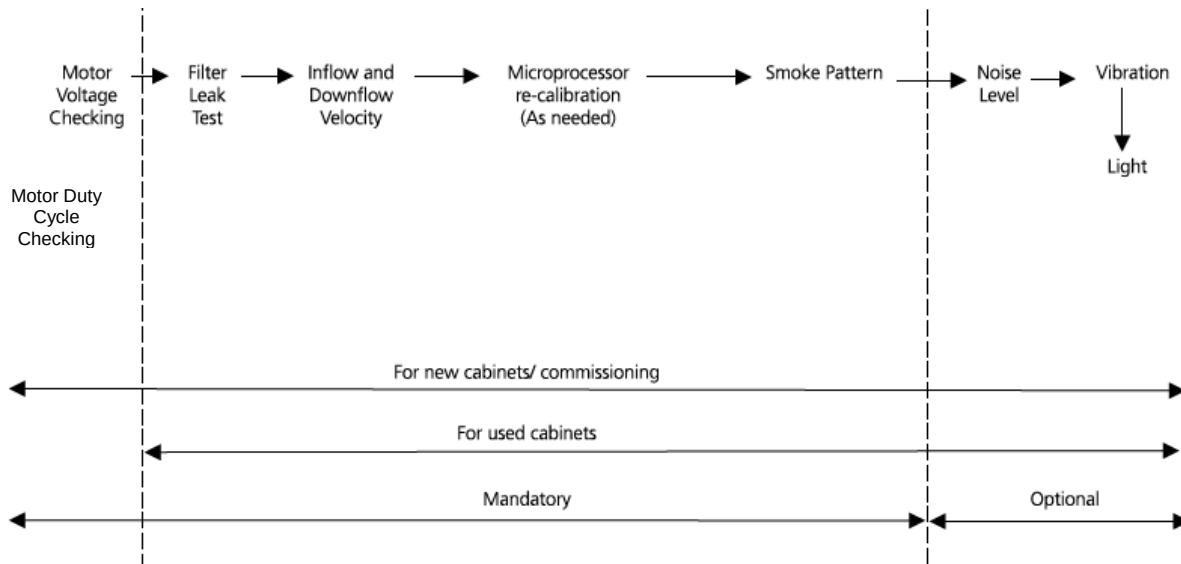
1. Check that the cabinet was installed away from air conditioning unit, heater, door, ceiling lamp, or any other equipment that affect cabinet airflow to ensure the sensor reading on top of cabinet is undisturbed.
2. Make sure the holes on the exhaust airflow sensor are not blocked, and the holes on the exhaust and downflow airflow sensors are perpendicular to the airflow path. If the airflow sensor is moved, then the microprocessor requires re-calibration.
3. Make sure the holes on the exhaust and downflow airflow sensors tube are perpendicular to the airflow path. If the airflow sensor tube is moved, then the microprocessor requires re-calibration.
4. Make sure that the exhaust filter is not blocked, and the downflow diffuser is properly installed.
5. Make sure that the downflow diffuser is properly installed.
6. Check that the fan duty cycle matches with the value given on the test report.

1.2. Cabinet Field Certification

Cabinet field certification must be performed on:

- New installation.
- Relocation of cabinet.
- Activation of airflow alarm with undetermined cause.
- When cabinet performance is suspect.
- At least once in a year or based on your risk assessment.

The steps to perform field certification:



1.3. Performing Calibration / Certification

1.3.1. Setting Cabinet in Maintenance Mode

To start the certification, ensure the fan and lamps are turned on. The cabinet should then be put to maintenance mode. In Maintenance Mode all alarms are defeated and the window can be moved without the light being switched off.

1. Press MENU button. Enter the ADMIN PIN (Default ADMIN PIN is 0009)
2. If SETTINGS is displayed the ADMIN code has been removed proceed to step 3
 - a. Press SET with 0 flashing
 - b. Press SET with 00 last digit flashing
 - c. Press SET with 000 last digit flashing
 - d. Press SET with 0000 last digit flashing
 - e. Press ▼ button to key in 9
 - f. Press SET
3. Alarm will sound, wait until alarm stops. Cabinet should read.

```

MENU OPTIONS XX:XX
→ SETTINGS
  SET MODE
  FIELD CALIB
  ADMIN SETTINGS
  
```

4. Use the ▼ button until the LCD displays SET MODE. Press the SET button and LCD would show :

```

MENU OPTIONS XX:XX
→ NORMAL MODE
  QUICKSTART MODE
  MAINTENANCE MODE
  
```

5. Use the ▼ button until the LCD displays MAINTENANCE message at the top most line of the LCD. Press the SET button. The cabinet is now in maintenance mode and the word "Maintenance" will blink continuously to indicate unsafe condition.
6. Press the MENU button twice to exit from the menu tree.

7. The word "MAINTENANCE" is displayed on the LCD as shown below.

MAINTENANCE XX:XX
Sash: OK
I: 0.53 m/s D: 0.35 m/s

OR

MAINTENANCE XX:XX
Sash: OK
I: 105 fpm D: 70 fpm

Note: Downflow velocity for 6ft unit is 0.33 m/s (65 fpm)

1.3.2. Locating the Speed Controller

Prepare for certification by locating the speed controller which you would need to use for changing the motor speed.

1. Remove the two screws holding the front panel. (One screw located on the blue panel at each end on bottom edge).
2. Lift front panel. It will be held open by the gas spring.
3. Locate speed controller positioned toward right side of electrical panel.

1.3.3. Inflow Velocity Test

1.3.3.1. Direct Inflow Measurement (DIM)

Direct Inflow Measurement (DIM) is the most accurate method to measure inflow. It is to be performed using an 8" flow hood that measured the inlet volumetric flow rate on the front aperture.

1. Attach the flow hood inlet to the front aperture, with the window lowered to touch the flow hood and the rest of the opening area covered by laminated paper and masking tape.
2. Take 5 readings and divide the volumetric inflow reading in liter/second by the opening area to get the inflow velocity in meter/second. The opening dimension (width x sash height), and the required nominal inflow velocity of various LA2 cabinets is as follows:

Cabinet	LA2-3Ax	LA2-4Ax	LA2-5Ax	LA2-6Ax
Target Nominal Inflow Point (l/s or cfm)	118 (251)	154 (329)	191 (406)	201 (429)
Target Fail Inflow Point (l/s or cfm)	89 (189)	116 (247)	144 (306)	152 (323)
Target Nominal Inflow Point (cmh)	424	555	686	724
Width (m or ft)	0.970 (3)	1.270 (4)	1.570 (5)	1.870 (6)
Sash Height (m or ft)	0.229 (3/4)	0.229 (3/4)	0.229 (3/4)	0.203 (11/16)
Aperture Area (A) (m ² or ft ²): width x height	0.222 (2.391)	0.291 (3.130)	0.360 (3.870)	0.380 (4.086)

3. Remove the flow hood.

1.3.3.2. Secondary Inflow Measurement

In the event that flow hood is not available, a secondary inflow method to measure inflow can be performed.

1. Remove armrest, but install it back after testing. Hold the armrest by hand when the last capped nut is unscrewed.
2. Put the masking tape to cover the small gap between the tray and the cabinet. Mark the proper probe position on the masking tape as below (for example):

Cabinet	LA2-3Ax	LA2-4Ax	LA2-5Ax	LA2-6Ax
Width (mm) (ft)	970 (3)	1270 (4)	1570 (5)	1870 (6)
Sash height (mm) (in)	229 (9)	229 (9)	229 (9)	203 (8)
Distance from walls (mm) (in)	121.3 (4 3/4)	127.0 (5)	130.8 (5 1/8)	133.6 (5 1/4)
Distance apart (mm) (in)	121.3 (4 3/4)	127.0 (5)	130.8 (5 1/8)	133.6 (5 1/4)

- Put the gage block and lower the sash window until it consistently touches the window height template on the left and right side of the cabinet. REMOVE the gage block before taking airflow reading.
- Mount the thermo-anemometer on the probe holder, which along with gage block, may be obtained by contacting biotech@escoglobal.com. Attach the probe holder to the cabinet by making the magnets on the probe holder bonds to the EG steel part underneath the cabinet. Make sure the probe holder touch the front nosing of the cabinet. Adjust the probe height so that the center of the white tip is 38 mm (1.5 inches) below the sash window lower edge to take the reading at the center of the opening height.
- Set the thermo-anemometer to 20 sec averaging function. Take the readings and record the inflow velocity below:

LA2-3 (7 points)												
LA2-4 (9 points)												
LA2-5 (11 points)												
LA2-6 (13 points)												
1	2	3	4	5	6	<u>7</u>	8	<u>9</u>	10	<u>11</u>	12	<u>13</u>

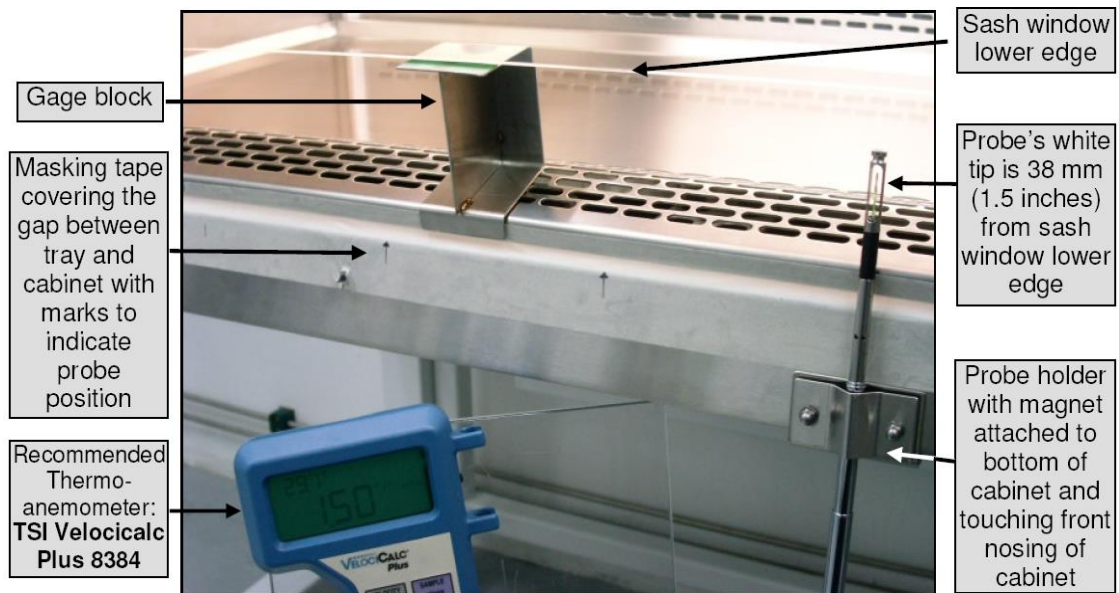
- Take the average of the readings. Multiply the average by the sash height conversion factor:

Cabinet	LA2-3Ax	LA2-4Ax	LA2-5Ax	LA2-6Ax
Conversion factor	0.324	0.336	0.313	0.342

Example:

If the average reading of LA2-4A at 76 mm (3 inches) opening is 1.58 m/s (313 fpm) then the inflow reading at normal window opening is: $1.58 \text{ m/s} \times 0.336 = 0.53 \text{ m/s}$ or $313 \times 0.336 = 105 \text{ fpm}$

Note: Calibration must be done with sash window at normal opening, not 3" (76 mm) opening



Recommended instrument: TSI Inc. (USA) 8384 Thermo-Anemometer (auto temperature correction)

Accuracy: $\pm 3\%$ of reading.

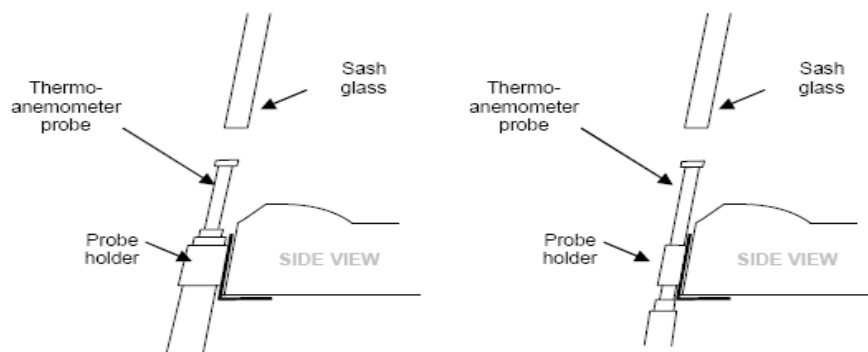
**only for field certifier:*

Average Inflow reading at 76mm (3 inches) : m/s | fpm

Inflow Reading Converted to Normal Sash Opening : m/s | fpm

Result : Pass / Fail

Note: It is very important that the orientation of the probe with respect to the sash glass be correct or the inflow measurement will not be accurate. The probe must be parallel to the sash glass angle and in the same plane as the sash glass. Adjust probe as required to maintain plane of window

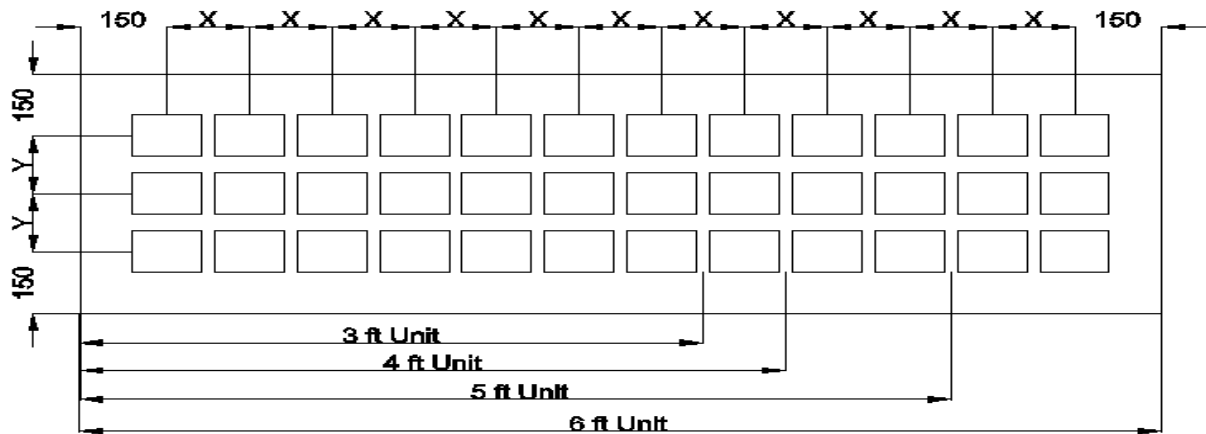


If the sash glass is too far forward clamp the thickest part of the probe with the holder to bring the probe forward to be inline with the glass

If the sash glass is too far back clamp the thinnest part of the probe with the holder to move the probe towards the back and inline with the glass

1.3.4. Downflow Velocity Test

Downflow testing is performed using thermo-anemometer on a plane that is 10 cm above the lower edge of the sash window opening for the cabinet following the grid below:



Note: The velocity probe must be placed accurately within 12.5mm (0.5 inch) of the above grid, with correct orientation (see [Probe Orientation](#) note)

Cabinet		LA2-3Ax	LA2-4Ax	LA2-5Ax	LA2-6Ax
Width (mm) (ft)		970 (3)	1270 (4)	1570 (5)	1870(6)
Left to Right	Distance from walls = a (mm) (in)	150 (6)	150 (6)	150 (6)	150 (6)
	Distance apart = X (mm) (in)	110 (4 5/16)	139 (5 1/2)	141 (5 9/16)	143 (5 5/8)
Front to Back	Distance from walls = b (mm) (in)	150 (6)	150 (6)	150 (6)	150 (6)
	Distance apart = Y (mm) (in)	140 (5 1/2)	140 (5 1/2)	140 (5 1/2)	140 (5 1/2)
Nominal Downflow (m/s) (fpm)		0.35 (70)	0.35 (70)	0.35 (70)	0.33 (65)

Acceptance : Average downflow velocity for 3, 4 and 5 ft cabinets: **0.35 m/s** $\pm$ 0.025 m/s or **70 fpm** $\pm$ 5 fpm

Average downflow velocity for 6 ft cabinet: **0.33 m/s** $\pm$ 0.025 m/s or **65 fpm** $\pm$ 5 fpm

All individual downflow velocity readings must be within $\pm$ 0.08 m/s (16 fpm) of the average or $\pm$ 20% of the average, whichever is greater

Note: Probe Orientation for downflow velocity measurements:

- ★ Front row - probe pointed to the front
- ★ Middle and back row - probe pointed to the right side wall on the right half of the cabinet, and pointed to the left side wall on the left half
- ★ If there is a reading located side-by-side in the center of the back or middle row the probe should be pointed toward the left side wall

1.3.5. Balancing Inflow and Downflow Velocity (If Required)

The proper inflow and downflow velocity is achieved by balancing the motor duty cycle and exhaust damper located on top of the cabinet as required.

The damper is used to balance the downflow velocity and exhaust velocity. The damper needs to be adjusted when the downflow velocity does not meet the accepted criteria (3', 4', 5' unit: 0.35 m/s and 6' unit: 0.33 m/s). Open the damper opening if the downflow velocity is too high; close the damper opening if the downflow velocity is too low.

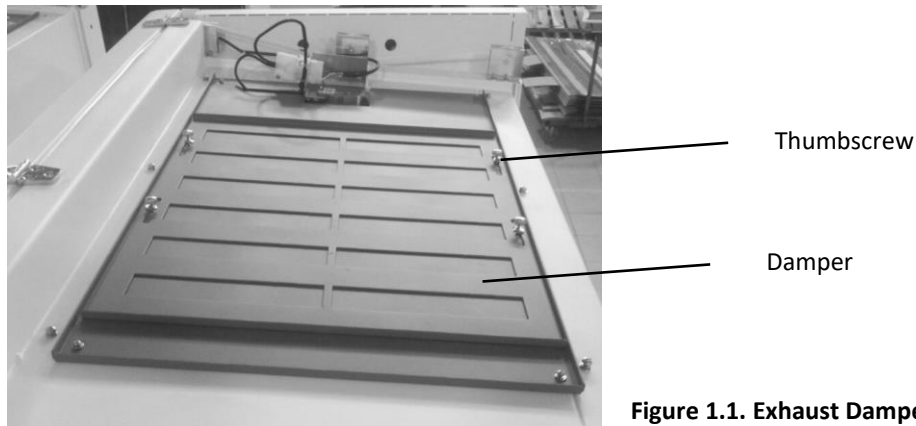


Figure 1.1. Exhaust Damper

Loosen the thumbscrew(s) to slide the damper left or right to increase or reduce the downflow velocity.

1.3.6. Calibrating the Microprocessor (Not Required on New Cabinets)

Calibrating the microprocessor establishes correlation between the sensor analog voltage output and the airflow measured with an independent instrumentation. This enables the microprocessor to display airflow velocities accurately. Calibration is required if the display does not correspond 0.02 m/s (± 4 fpm) with the instruments used to certify the cabinet.

There are **2 steps** to do standard field calibration:

A. Balance the Motor Duty Cycle and Damper Opening to get Nominal Inflow and Downflow

B. Calibrate the Nominal Inflow and Downflow Set Point

Contact Esco for special calibration procedure when Airflow sensor, Main board, or Software is replaced.

A. Balance the Motor Duty Cycle and Damper Opening to get Nominal Inflow and Downflow

You need the duty cycle that correlate to the Nominal Inflow and Downflow. This is a prelude to calibrating the microprocessor.

1. If the cabinet airflow has not been properly balanced, place the Thermo-Anemometer in the work area to measure the downflow, then calculate the average downflow (3, 4, 5 feet Unit should be 0.35 m/s $\pm$ 0.025 m/s (70 fpm $\pm$ 5 fpm) while 6 feet Unit should be 0.33 m/s $\pm$ 0.025 m/s (65 fpm $\pm$ 5 fpm)). Adjust the motor duty cycle and adjust damper opening as necessary (See adjustment table).
2. If the cabinet airflow has not been properly balanced, place the Flow Hood on the front of the cabinet to ensure correct Nominal Inflow (0.53 m/s $\pm$ 0.025 m/s or 105 fpm $\pm$ 5 fpm). Adjust the motor duty cycle and adjust damper opening as necessary (See adjustment table).

Adjustment Table

IF			THEN	
Inflow	Downflow		Motor Duty Cycle	Damper Opening
-	-	➔	+	Constant
+	+	➔	-	Constant
+	-	➔	Constant	-
-	+	➔	Constant	+
++	-	➔	-	-
-	++	➔	-	+
+	--	➔	+	-
--	-	➔	+	+
Legend: + = Strong ++ = Very Strong - = Weak -- = Very Weak			Legend: + = Increase - = Decrease	

B. Calibrate the Nominal Inflow and Downflow Set Point

No	Action	LCD Display
1.	Press the MENU button. No PIN is asked if cabinet is in maintenance mode. Otherwise, please key in the ADMIN PIN. Press the ▼ button to select FIELD CALIB, then press SET.	MENU OPTIONS 17:23 SETTINGS SET MODE ➔ FIELD CALIB
2.	The pointer will be at Airflow Calib. Press SET.	MENU OPTIONS 17:23 ➔ Airflow Calib Reset Calibration
3.	The "Read Manual" message is to discourage unauthorized person. Press SET to continue.	CALIBRATION 17:23 Field Calibration Read Manual!!! and Press SET Button
4.	From previous Step A , ensure that you have balanced the airflow at: • Inflow: $0.53 \text{ m/s} \pm 0.025 \text{ m/s}$ (105 fpm $\pm 5 \text{ fpm}$) • Downflow: $0.35 \text{ m/s} \pm 0.025 \text{ m/s}$ (70 fpm $\pm 5 \text{ fpm}$) (Except LA2-6A_: $0.33 \text{ m/s} \pm 0.025 \text{ m/s}$ (65 fpm $\pm 5 \text{ fpm}$)) Then press SET.	CALIBRATION 17:23 Set Inflow 0.53 m/s And Downflow 0.35 m/s and Press SET... OR CALIBRATION 17:23 Set Inflow 105 fpm And Downflow 0.35 m/s and Press SET...
5.	The Inflow velocity is blinking meaning it can be changed. Select your obtained Inflow using ▲ or ▼ buttons then Press SET.	CALIBRATION 17:23 Enter Inflow: 0.53 m/s Use ▲/▼ to change Then Press SET... OR CALIBRATION 17:23 Enter Inflow: 105 fpm Use ▲/▼ to change Then Press SET...
6.	Wait for 3 minutes for the software to record this sensor ADC point. Do not change motor duty cycle or disrupt airflow / sensor during this time	CALIBRATION 17:23 Wait 03min Time Left: 02:59
7.	The Downflow velocity is blinking meaning it can be changed. After 3 minutes, software will ask you to enter your obtained Downflow. Select your obtained Downflow using ▲ or ▼ buttons. You can immediately Press SET.	CALIBRATION 17:23 Enter Downflow velocity: 0.35 m/s Then Press SET... OR

No	Action	LCD Display
		CALIBRATION 17:23 Enter Downflow velocity: 70 fpm Then Press SET...
8	Calibration Done message will be displayed, and the screen will go back to MENU OPTIONS like on Step 1. If Calibration Error message is displayed, please ensure that Inflow and Downflow are within ± 0.25 m/s from nominal, and re-do the calibration.	CALIBRATION 17:23 Enter Downflow velocity: 70 fpm CALIBRATION DONE
9	Return the cabinet from Maintenance Mode to Normal Mode.	

MICROPROCESSOR CALIBRATION IS FINISHED

Setting the Half Speed Inflow

1. Set the cabinet to Half Speed Mode. Press black bottom on the ECMS Controller a few second until reading "0".
2. Adjust the Half Speed Duty cycle to get the half speed inflow **0.37 m/s** or **73 fpm**.
3. Press black bottom on the ECMS Controller a few second until reading "dn".
4. Set the Duty Cycle to Normal Speed Mode.

1.3.7. Motor Control Verification

The airflow compensation feature of the motor should be verified by the following methods.

1. Ensure the cabinet airflow is now balanced.
2. Record the initial average inflow velocity and average downflow velocity.
3. Lift up the front panel to enable you to observe the motor duty cycle. Record the initial motor duty cycle.
4. Measure and record power consumption. Power consumption record is used as it reflects the need for the motor to draw more power to produce the required airflow as filter is loaded.
5. Close the front panel.
6. Block 50% of the inflow grill for 2 minutes and record the average inflow velocity, average downflow velocity and power consumption.
7. Unblock the inflow grill and let the cabinet stabilize for 2 minutes. Record the final average inflow velocity and average downflow velocity.

1.3.8. Airflow Smoke Patterns Test

A suitable visible smoke generator was used to visualize cabinet airflow patterns.

Downflow:

- Spray smoke on work surface center line, 10 cm (4") above sash window lower edge, left to right.
- Observation: Smoke flow should be smooth downward with no dead spots or reflux (upward blow).

View Screen Retention:

- Spray smoke 2.5 cm (1") behind sash, 15 cm (6") above sash lower edge, left to right.
- Observation: Smoke flow should be smooth downward with no dead spots, reflux (upward blow) or escape.

Work Opening Edge Retention:

- Spray smoke along entire aperture perimeter, 3.8 cm (1.5") outside cabinet.
- Observation: Smoke flow smooth, no reflux out once drawn in, no billowing or penetration over work surface.

Sash Wiper Seal:

- Spray smoke behind sash, 5 cm (2") from the sides and along the top of the work area.
- Observation: There should be no smoke escape from the cabinet.

Downflow Diffuser Perimeter:

- Spray smoke at 2.5 cm (1") underneath the diffuser perimeter (this is a non-NSF test).
- Observation: Smoke flow should be downward with no dead air corner or reflux.

Notes:

The sash Wiper Seal Test will require two persons to perform the test:

- One person to insert their hand into the cabinet and eject the smoke.
- One person to check for smoke escape from the cabinet, with the front panel lifted.

1.3.9. ADDENDUM: Pressure Retention Test (Soap Bubble Test)

The purpose of pressure retention test is to look for possible leakage through the exterior joints, welds, silicon sealants, gaskets, and plenum penetrations as depicted in Figure 1 and Figure 2 below.

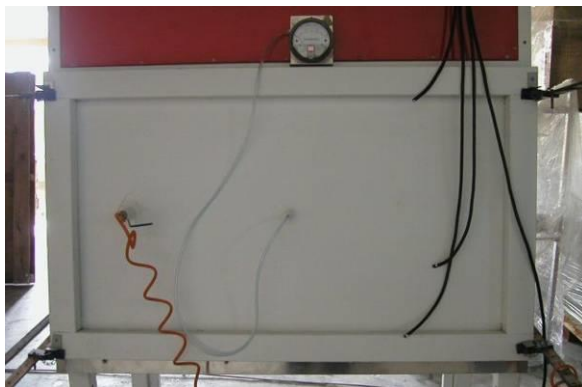


Figure 1.2 Front Cover



Figure 1.3 Top Cover

First, the sash track cover must be removed and the sash window is lifted. There is no need to cut any electrical wires.

The front opening is closed by a panel (Figure 1), as well as the top ULPA filter opening (Figure 2).

Please contact your Esco sales representative to order the Pressure Test Kit should this test need to be performed on-site.

The front cover panel has two ports (Figure 1): one port for the pressurized air tubing with valve, and the second port are connected to a pressure gauge.

The sealed cabinet is pressurized slowly to 500 Pa (2 " WG), and then the pressurizing air is turned off.

In 10 minutes check the pressure reading, should not be less than 375 Pa (1.5 "WG).

A drastic reduce in pressure (less than 1.5 "WG in 10 minutes) indicates that there is big leakage on the cabinet.

If the pressure after 10 minutes is less than 375 Pa (1.5 "WG), increase the air pressure back to 500 Pa (2 " WG) and maintain it. Soap solution is applied with a brush along the exterior joints, silicon seal, gaskets, and plenum penetrations to find the leakage indicated by soap bubbles.

Patch the leak with silicone and re-do pressure retention test.

Acceptance: With initial pressure of 500 Pa (2 "WG), the pressure after 10 minutes should not be less than 375 Pa (1.5 "WG)

Acceptance criteria / testing procedures have been adopted in accordance with the requirements of the:

- NSF/ANSI 49:2011 Standard (Section 6.2 and Annex A Section 1)

Note: All excess of soap solutions must be wiped off immediately after testing, to prevent stain/corrosion that might appear as the effect.

1.3.10. Mandatory Annual Inspection of Sash Mechanism (For Motorized Sash BSC)

Checking of Motorized Tubular Motor:

1. Remove screws from the left and right side of blue panel. Lift up the front panel.
Caution: The front panel is equipped with a gas spring which will open the front panel automatically.
2. Check mounting screws on both sides of the brackets, ensure there are no loose screws.
3. Check the plastic end caps for any sign of cracks. Do not operate sash if cracks are found.
4. Check suspension belts are securely fastened to the sash glass. Check for any sign of tearing on the belt.

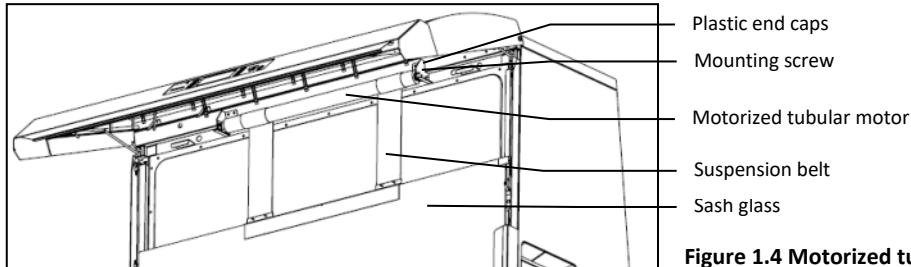


Figure 1.4 Motorized tubular motor

Checking of the Sliding Sash Mechanism

1. Remove screws from the left and right side of the sash cover.
2. Remove the sash covers for access to the sliding window mechanism.
3. Ensure Guide A is securely glued to the glass.
4. Check by lifting the glass out slightly and make sure Guide A does not detached from glass with finger force.
5. Check for the gap between the sash guides as per sketches shown above.
6. Check the position of magnetic switch to ensure it does not collide with the moving sash glass.
7. After all the above checks are done, install the sash cover and close the front panel.
8. Operate the sash and the glass should be moving smoothly up and down.
9. Ensure that the glass does not jam at any position and the suspension belt does not slack at any point in time.

1.4. Parts Replacement

The use of non-Esco parts and / or parts not supplied directly by Esco or our authorized distributors, including but not limited to maintenance parts, spare parts, replacement parts, system components and / or system accessories, shall void all expressed or implied warranties.

1.4.1. Filter Replacement Procedure

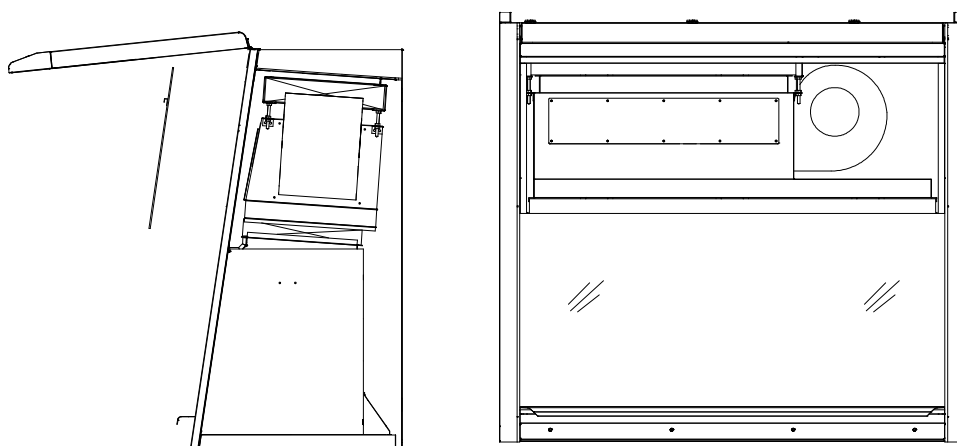


Figure 1.5 Filter Replacement

The filters should not be replaced until the entire cabinet has been decontaminated.

1. Remove the screws from the bottom of blue panel and then lift up the front panel.

Caution: The front panel is equipped with gas spring which will open the front panel automatically.

2. Move sash window to the lowest position.

Note: For motorized sash BSC, remove the window straps before moving the sash window to the lowest position.

3. Remove red plenum cover using Phillips screwdriver.
4. Remove the cover on the positive plenum box.
5. Using a #13 spanner, loosen the nut until the exhaust telescoping frame collapses all the way down along with the exhaust filter. Please ensure the loosening is done slowly, one nut at a time.
6. Remove exhaust filter by sliding out from front.
7. Using the #13 spanner, tighten the nut until the positive air plenum box is lifted up and the downflow filter can be pulled out.
8. Remove the downflow filter by sliding out from the front.
9. To replace filter, repeat the above mentioned procedure in reverse order

Warning: two persons may be required for changing the filter(s) of 5 ft and 6 ft cabinets

The cabinet must undergo re-certification after changing filter.

Under normal usage and barring any accident (example - puncture), the two ULPA filters do not need replacement unless and until the inflow and / or downflow velocity can no longer be maintained within the specifications.

1. Before the filters are removed from the cabinet, their contaminated faces should be taped off using plastic cardboard or some other suitable material. This would minimize the risk to personnel in the event of decontamination not being thorough.
2. Used filters should be disposed of in accordance with local regulations. They may have to be incinerated as medical waste. They should be double bagged and appropriately labeled after being removed from the cabinet.
3. Proper PPE (based on your risk assessment) should be worn while removing used filters. You should dispose of these things after the procedure. Finally you should properly wash your hands.
4. Before the new filters are installed, all surfaces should be thoroughly cleaned of silicon and/or adherent gasket material. The new filter should be carefully handled and examined before being fitted. It is important that the filters and the gaskets be checked for leaks prior to use.

1.4.2. Blower Replacement Procedure

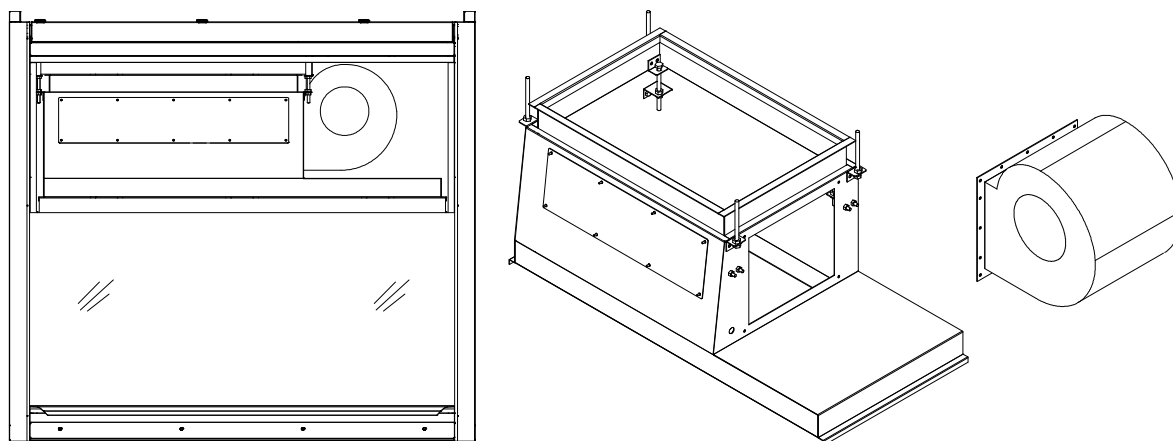


Figure 1.6 Blower Replacement

To replace the blower, follow the steps mentioned below:

1. Disconnect the electrical connection.
2. Open the front panel.
3. Open the red cover.
4. Unscrew the blower housing.
5. Take out the blower housing and unscrew the bolts that secure the blower to the blower housing.
6. Repeat this procedure in reverse order with the new blower to complete the replacement.

1.4.3. Airflow Sensor Replacement Procedure (Doughnut Sensor Housing: LA2-E)



Figure 1.7. Air Flow sensor mounting

1. Airflow sensor is located on top of the cabinet. Start by disconnecting the airflow sensor cable connector.



Figure 1.8. Disconnecting Airflow sensor Cable

2. Remove temperature sensor from the bracket.



Figure 1.9. Remove Temperature sensor

3. Remove the doughnut sensor housing from the bracket.

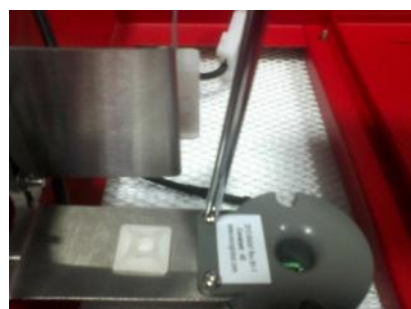


Figure 1.10. Remove doughnut sensor housing

4. Install the new sensor (it comes with the doughnut sensor housing) into the bracket and connect the sensor cable connector. Ensure the airflow direction described in the sensor housing is complied with during installation.



Figure 1.11. Airflow sensor direction

1.4.4. Sash Glass Replacement Procedure

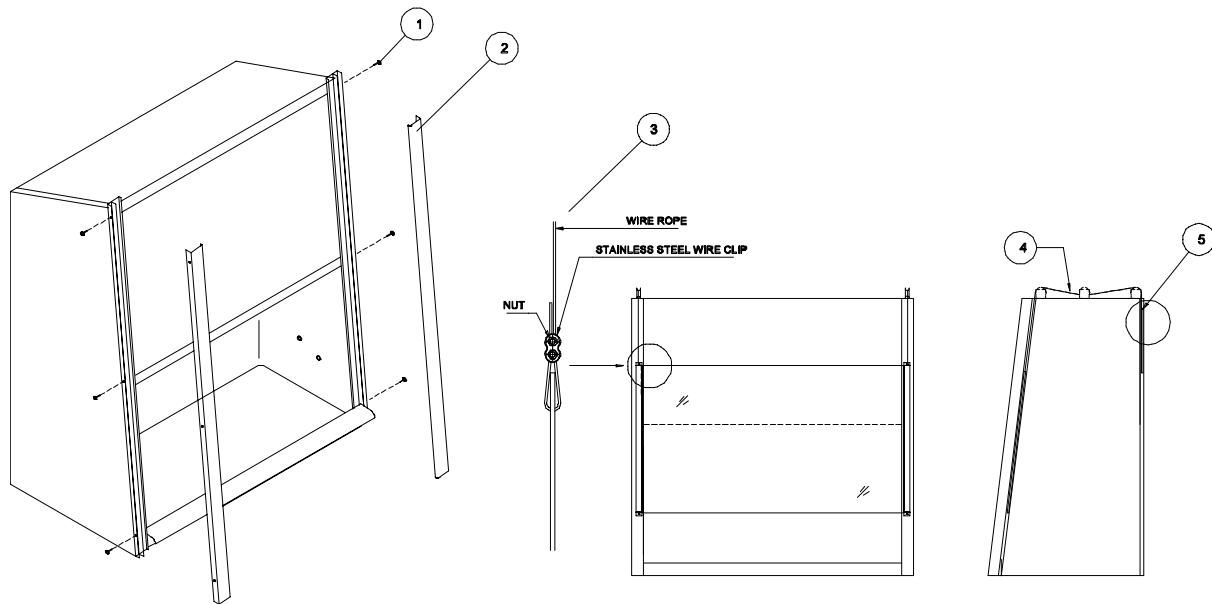


Figure 1.12. Sash Glass Replacement

1. Remove the screws under the blue panel to open the front panel.
2. Remove the countersunk screws from the side front profile (1). Open both profile covers (2).
3. Bring the sash glass to fully closed position that it is resting on the armrest.
4. Support the counterbalance at the back on its highest position by using a wooden bar or stacker. Make sure that the counterbalance is well supported before disconnecting the wire from the sash glass. Do not hold with hand, as the heavy counterbalance can slip down easily and injure the holder.
5. Disconnect the glass from the wires by loosening the screws on the clips (no. eight shape clips) (3).
6. Replace the glass.
7. Hook the wire across the 3 rollers (4), Run the wire back to the hole (5) and secure it with the clips (3). Ensure the clips are perfectly tightened. Observe the position of where the wire is previously clipped. This will help to ensure that the sash wire is connected to counterbalance evenly.
8. Remove the wood supporting the counterbalance at the back to allow counterbalance to move.
9. Move the sash glass to test whether the connection is tight.
10. Place back the profile covers and secure it.

1.4.5. Fluorescent Lamp(s) Replacement Procedure

1. Disconnect the electrical connections.
2. Unfasten the two screws from the bottom of the blue panel and then lift up the blue panel.
Caution: The front panel is equipped with a gas spring which will open the front panel automatically.
3. The fluorescent lamp(s) is located behind the blue panel. Remove fastening ties from fluorescent lamp
4. Disconnect the old one(s) and replace with the new one(s).

1.4.6. UV Lamp Replacement Procedure

1. Disconnect the electrical connections.
2. Take out the old UV lamp by rotating it 90° counter clockwise and pull downward.
3. Install the new UV lamp by pushing it upward and rotating it 90° clockwise to secure the lamp in its socket.

1.4.7. Service Fixture Plumbing

In order to replace, repair, or install a plumbing system, following steps need to be followed:

1. Open the front panel and keep it at fully open position.
2. Open the front profile cover on the side where the plumbing is, by loosening the countersunk screws (A) from the side of the cabinet.

3. Unscrew 3 countersunk screws that connect the profile base to the side dress panel (B) and screws that connect the module to the dress panel at the back (C).
4. Lift the dress panel slightly up (around 15 mm), then pull the dress panel out to make the fixtures provision exposed.
5. Replace/ Repair/ Install the plumbing while it is open. Ensure connection is tight and Loctite/seal tape is being used to prevent leakage. Test the connections based on Esco pre-installed piping standard of 80 psi, and check whether there is leakage or not.
6. Place back the side dress panel and reverse steps a to d to complete the installation.

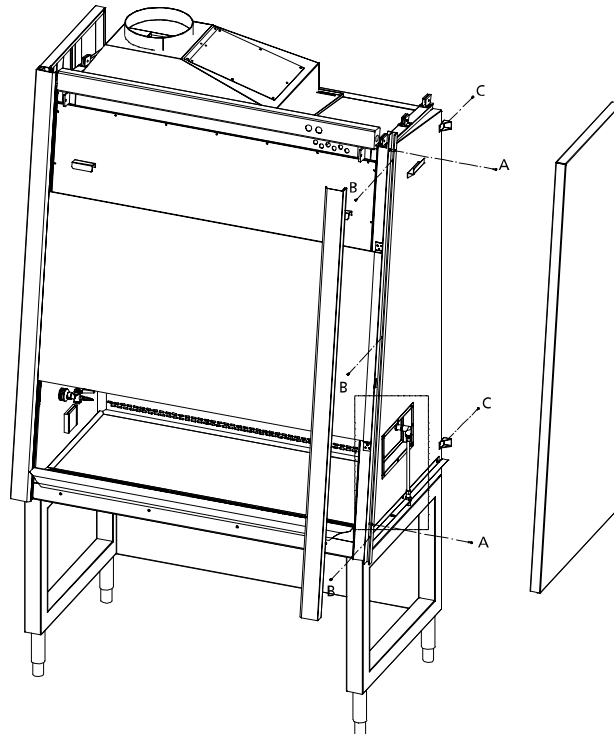


Figure 1.13. Service Fixture Plumbing

Chapter 2 - Decontamination

BSC decontamination should be performed in any of the following eventualities:

- Before replacing the exhaust/downflow filters or blower.
- Before accessing the contaminated negative or positive plenum of the BSC.
- In case of an accidental spillage that might have contaminated any of the inaccessible surfaces.
- Before performance validation and re-certification.
- Before BSC relocation.
- Before any procedures mandated by user's risk assessment.

Appropriate Personal Protective Equipment (PPE) per user's risk assessment that may include gloves, gown, and respirator with filter suitable for the decontamination gas/vapor must also be used by the certifier or any other qualified personnel performing the decontamination.

Esco can provide the Formalin Vaporizer (FV-001) and decontamination bag needed for the decontamination procedure.

The information provided in this section does not construe a final advice on how decontamination must be carried out. Appropriate personnel including the safety personnel in the facility must be consulted and adequate risk assessment must be conducted to ensure the safety and efficacy of the decontamination procedures.

2.1. Decontamination Agents

2.1.1. Formalin/Paraformaldehyde Decontamination

Typically the decontamination is performed using formalin gas by either vaporizing 37% formalin solution or by de-polymerization of solid paraformaldehyde.

Despite its widespread usage for decontamination, formalin gas presents the following health risks:

- External contact can cause irritation to skin, eyes, and mucous membranes.
- Inhalation in small concentrations can cause coughing, nausea, and diarrhea.
- Inhalation in large concentrations can cause convulsions, coma, and death.
- Long term exposure can cause cancer.

Although the Permissible Exposure Level (PEL) for formalin is 0.75 ppm, many scientists believe that there is no safe level of carcinogen exposure to humans. Therefore, typically the room must be evacuated when the decontamination process is performed, which leads to laboratory down time.

The use of formalin decontamination also has other disadvantages:

- The process is time-consuming.
- The certifier needs to pulse the BSC fan to circulate the formalin vapor. This can dislodge the tape holding the plastic sheet covering the exhaust filter.
- Due to excessive residue extensive cleaning must be done after decontamination and before use.

According to OSHA [Occupational Safety and Health Administration (USA)], formaldehyde Short Term Exposure Level (STEL) is 2 ppm for 15 minutes exposure, 4 times a day, minimum of 60 minutes in between exposures. Any additional local safety regulations should also be observed. Personnel should be given adequate training. The following links provide general guidelines on formaldehyde safety:

- Regulations (Standards - 29 CFR) Formaldehyde - 1910.1048, Occupational Safety and Health Standards, OSHA (Occupational Safety and Health Administration), U.S. Department of Labor: <http://www.osha.gov>
- OSHA Formaldehyde Fact sheet (PDF format). Available at: http://www.osha.gov/OshDoc/data_General_Facts/formaldehydefactsheet.pdf

The time required to carry out formalin decontamination is outlined below:

No.	Process	Time
1	Set-up & sealing the BSC to make it air tight	1 hour
2	Formalin vaporization	½ hour
3	Formalin contact time to obtain target log of 4-6 kill	8 – 10 hours
4	Ammonia vaporization to neutralize formalin	½ hour
5	Ammonia contact time to neutralize formalin	2 hours
6	Exhausting the ammonia residue	1 hour
7	Tear-down & cleaning the (substantial) residue	1 hour
TOTAL without ammonia neutralization		10½ – 12½ hours
TOTAL with ammonia neutralization		14 – 17 hours

Ammonia is used to neutralize formaldehyde. OSHA prescribes the ammonia STEL is 35 ppm for 15 minutes exposure, 4 times a day, minimum of 60 minutes in between exposure.

Due to the adverse health effect of formalin gas, its use has been banned in some countries such as Germany, Austria, and Switzerland. Other European countries are expected to follow suit. Two primary candidates to replace formalin decontamination are chlorine dioxide gas and hydrogen peroxide vapor.

2.1.2. Chlorine Dioxide Decontamination

Chlorine dioxide decontamination is performed by injecting chlorine gas (Cl_2) into a cylinder filled with solid sodium chlorite (NaClO_2), which generates the greenish-yellow chloride dioxide gas (ClO_2). Chlorine dioxide decontamination is much faster than formalin. Being a true gas, it spreads quickly, without the need of pulsing the BSC's blower. It can rapidly kill the micro-organisms with high efficacy with just 1 hour contact time. User must note that the concentration of chlorine gas inside the cabinet must decrease to a safe level (e.g. by using scrubber system) before user should open the air tight seal. The Short-Term Permissible Exposure Limit (STEL) for Chlorine Dioxide gas is 0.3 ppm. The time required for the entire process of chlorine dioxide decontamination is as follows:

No.	Process	Time
1	Set-up & sealing the BSC to make it air tight	1 hour
2	Chlorine dioxide gassing	½ hour
3	Chlorine dioxide contact time	1 hour
4	Chlorine dioxide "scrubbing"	½ hour
5	Tear-down & cleaning the (minimal) residue	½ hour
TOTAL		3½ hours

Chlorine dioxide has the PEL of 0.1 ppm, compared to 0.75 ppm for formalin. In both processes airtight BSC sealing is required to protect personnel from the gas exposure.

1.1.2.1 Recommended Decontamination Sealing Method

This section gives an example of how the cabinet can be sealed when FV-011 formalin vaporizer is used. However, similar principle can be applied for other gaseous decontamination method.



1. Put the decontamination bag to enclose the entire BSC and support stand.



2. Use the 50 mm wide aluminum tape to seal the gap between the decontamination bag and the floor. To compensate for areas where the decontamination bag must turn, use multi section tape. Ensure that the two layers of tape overlap at the end points to prevent leakage.



3. Overlap the tape at the joints.



4. These 2 cables (for BSC and for vaporizer) should be combined so that only 1 cable should come out from the decontamination bag, to minimize chance of formalin leakage.



5. To seal the cable, first make an Ω -shape aluminum tape around the cable, and make a 5 cm flat portion going to both directions. Then, paste the aluminum tape to the floor.



6. Put the bottom part of the decontamination bag on top of the aluminum tape, and then seal it with another layer of aluminum tape. Use multiple layer of aluminum tape (*at least 2 layers*) on both directions to ensure leak tightness.



7. The BSC is ready for formalin decontamination.

2.1.3. Hydrogen Peroxide Decontamination

Hydrogen peroxide (H_2O_2) decontamination is performed by flash vaporization of an aqueous peroxide mixture, creating a vapor that is distributed throughout the inside the BSC.

STERIS and BIOQUELL are two major vendors of hydrogen peroxide generators. There are significant differences in operating principles.

- The STERIS principle is to avoid condensation on surfaces to minimize corrosion and optimize vapor distribution. The relative humidity inside the BSC must be lowered to 30% so that the remaining 70% relative humidity can be occupied by the hydrogen peroxide vapor.
- The BIOQUELL principle is to seek micro-condensation to achieve the kill. The generator releases tiny high-speed droplets inside the BSC.

Hydrogen peroxide vapor is non-carcinogenic, but highly effective against micro-organisms. Hydrogen peroxide (H_2O_2) vapor breaks down under catalytic action to become air and water, making it environmentally friendly and it leaves no residues. The BSC is often aerated by ducting to speed up the breaking down of hydrogen peroxide vapor.

The time needed for the entire process is outlined below:

No.	Process		Time	
1	Set-up & sealing the BSC		½ hour	
2	Conditioning and decontamination cycle		½ - 1 ½ hour	
3	Ducting out H ₂ O ₂	H ₂ O ₂ generator doing aeration	½ hour	8 hours
4	Tear-down		½ hour	
TOTAL			2 - 3 hrs	9½ - 10½ hrs

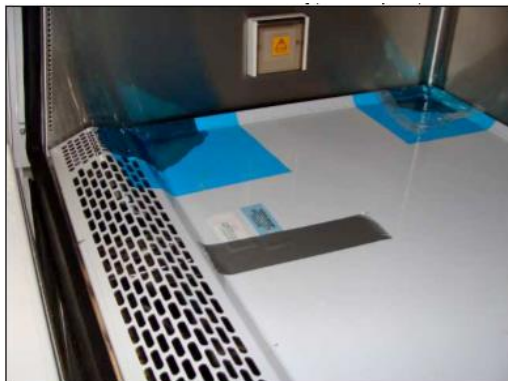
For hydrogen peroxide decontamination, the BSC need to be equipped with two ports:

1. One port located in front opening or side wall, penetrating the work zone area
2. One port located on top of the exhaust filter.
- 3.

The generator used defines the port function as described below:

	Steris	Bioquell
Hydrogen peroxide source	Injected into the BSC	Generated inside the BSC
Bottom front / side port	Hydrogen peroxide introduction	Hydrogen peroxide re-introduction
Top port	Hydrogen peroxide extraction	Hydrogen peroxide extraction

The Steps to perform the hydrogen peroxide decontamination with Bioquell:



1. Place the BIs on the work tray, drain pan, exhaust:



2. Place the internet catalyst and vaporizer on tray



3. Insert the hydrogen peroxide bottle:



4. Point the vaporizer fan towards front grille:



5. Cover exhaust filter with plate with 3" exhaust port:



6. Tape exhaust plate to cabinet. Connect 3" hose:



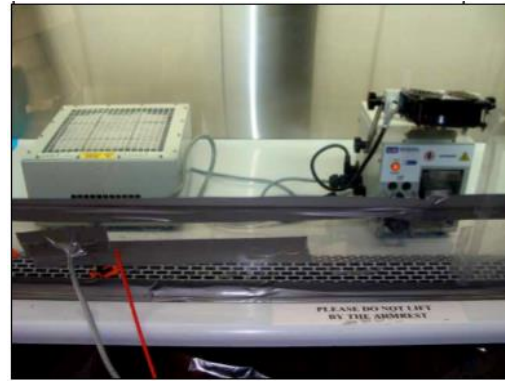
7. Connect 3" hose to 3" inlet on external catalyst:



8. Connect 1" hose to 1" outlet on external catalyst:



9. Open the small panel on side dress panel. Connect the 1" hose to port inside side dress panel:



10. Install front Perspex. Make control cable and pressure tube come out from holes. Seal with tape.



11. Put a wedge on left & right sash track cover:



12. Put the wedge behind the blue panel.



13. Adjust the valve at 3" outlet to get – 10 Pa:



14. Ensure the pressure in cabinet is around – 10 Pa:



15. Press 155 ml on controller despite we use 100 ml



16. The timer on control panel starts:



17. This is how the entire set-up looks like:



18. You may see condensation that then disappears:



19. Check the H_2O_2 concentration = 0 along tapes:



20. After gassing period ends, mute the alarm:



21. During aeration, set 3" outlet valve to full open, to increase the airflow and speed up the process:



22. During aeration, remove tapes at perspex holes to speed up aeration. Note that internal catalyst is on:



23. Check that H_2O_2 concentration = 0 inside cabinet by inserting the probe through the hole:



24. Remove Perspex, hoses, etc. Put BIs in soy broth and incubate for 48 hours at 37°C:

Check the results after 48 hours.

2.2. Further Information

- A Guide to Biosafety and Biological Safety Cabinets can be downloaded from <http://escoglobal.com/resource.php?id=13>
- An educational video on "Working Safely in your Biological Safety Cabinet" is available for viewing at <http://www.youtube.com/watch?v=ZnUW1N-JJz8>

Chapter 3 - Troubleshooting

3.1. Electrical and Mechanical Troubleshooting

This guide addresses the most common service issues. For more troubleshooting or service information contact your local Esco Distributor.

Hardware:

- DVM (Digital Voltage Meter).
Note: An analog meter can be used for troubleshooting, but cannot be used for motor voltage measurement.
- Phillips screwdriver.
- Insulated jumper cables.

The Electrical Panel and Component Layout:

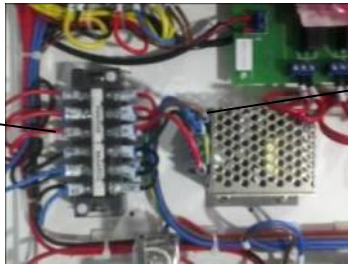
- Open the front panel, the electrical panel is located on the back of the panel. The component layout is as followed (please note that the layout is the same regardless of the power rating):

Problem 1: Cabinet does not start (LCD, button, fan light, and socket are inoperative)

Cause	Corrective Action
Power Failure	• Check wall outlet for power. • If the outlet has a switch make sure it is turned ON.
	Figure 3.1. Component Layout for LA2-3, 4, 5, 6A1, 2, 3-E Only
Power cord is not connected properly or faulty	• Ensure the power cord is connected correctly to the wall outlet. • Disconnect the cord at the cabinet and measure the AC voltage between the live and the neutral terminal of the cord. • If the voltage is not within $\pm 2\%$ of the wall socket voltage, replace cord; otherwise proceed to next step.

Circuit breaker has tripped	- Check the circuit breaker inside the electrical panel. NOTE: <i>If circuit breaker has tripped, do not reset the breaker before checking all electrical components and wiring connections (see Figure below).</i> - Does the cabinet operate correctly after resetting the circuit breaker? If not proceed to next step. Circuit breaker (normal condition) Circuit breaker (tripped condition) Figure 1-1d
Improper connection	- Measure AC voltage between LIVE (RED) and NEUTRAL (BLUE) terminal blocks inside electrical box (See Figure below). - Voltage must be within $\pm 10\%$ of the rated voltage. Refer to cabinet Serial Number Tag for required voltage. - If the voltage is out of the range or there is no voltage, check cable connection on both sides of connector A. 230 VAC Connector A pin 1 and 2 pin 3 is ground 115 VAC Connector A pin 1 and 2 pin 3 is ground - If you have voltage at Connector A, but not at the RED(LIVE) and BLUE(NEUTRAL) terminal block, the problem is between connector A and the terminal strips - If voltage is not present at connector A, BUT is present at the female connector for the line cord at the top of the cabinet that the power cord is plugged into the problem is between the female connector and the pins on Connector A. Figure 1-2
Defective power supply – Switch Mode Power supply (SMPS)	- Locate the SMPS (Power Supply) and Terminal block. - In the side of terminal block measure the DC voltage between Red (pin 3) and Black (pin 4) Measure on the side away from the SMPS. - The voltage should be in range of $+12\text{ VDC} \pm 10\%$. - If you do not have voltage, measures the other side of terminal block. - If out of range or no voltage, check incoming power to the SMPS – check terminal where the cord is connected. The input of SMPS should be $\pm 10\%$ of the cabinet incoming voltage. - If you have input voltage but no output voltage replace the SMPS. - If input and output voltage is correct proceed to next step.

SMPS output
Red-
Black/blue is
12 VDC



SMPS input Brown-
Blue-G/Y is 115 VAC or
230 VAC

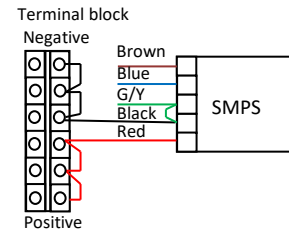


Figure 1-3

Connection
problem to
main board
or from
main board
to display

- Measure the incoming voltage on the main board at terminal J 13 (Note polarity, blue cable closest to edge is negative -).
 - Voltage should be between 10.8 –13.2VDC.
 - If voltage is out of range or no voltage, check connection between SMPS and main board.
 - Check flat ribbon cable connection from main board to LCD/keypad membrane.
- NOTE:** There are two ribbon cables from the main board and the display and key pad. The larger flat cable is for the Display. The smaller flat cable is for the LEDs/keypad membrane.
- If the LCD Display is on, but not the LEDs in keypad membrane, check connection on the smaller Flat Ribbon cable.
 - If incoming voltage at J13 is correct and flat ribbon cable connections are connected but the LEDs on the keypad membrane and LCD display are still not on proceed to next step.

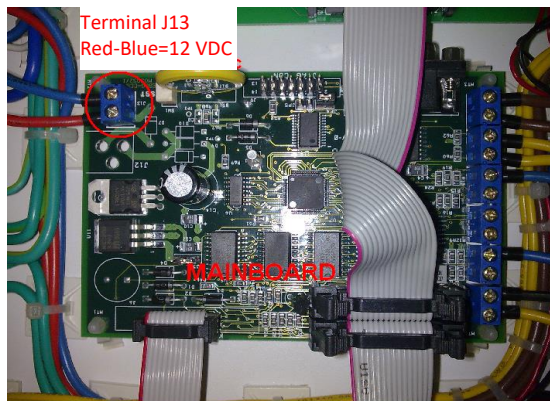


Figure 1-4

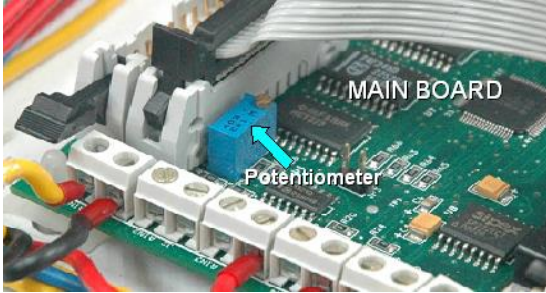
Defective
main board

- The main board is defective if the main board incoming supply is between 10.8 – 13.2 VDC and:
 - All LED's on the control panel are off.
 - The LCD is blank.
 - No buzzer sound.
 - If these conditions exist, replace the main board. Otherwise proceed to next step
- Note:** when replacing main board, reconnect all wires correctly, any wrong wiring may result in damage.



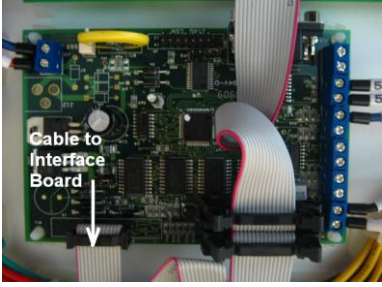
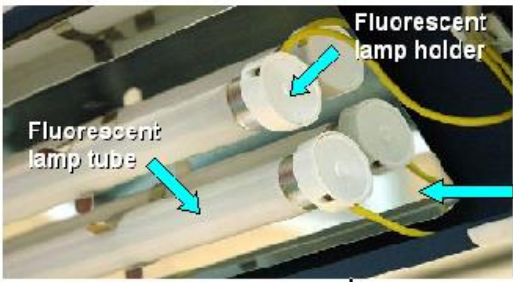
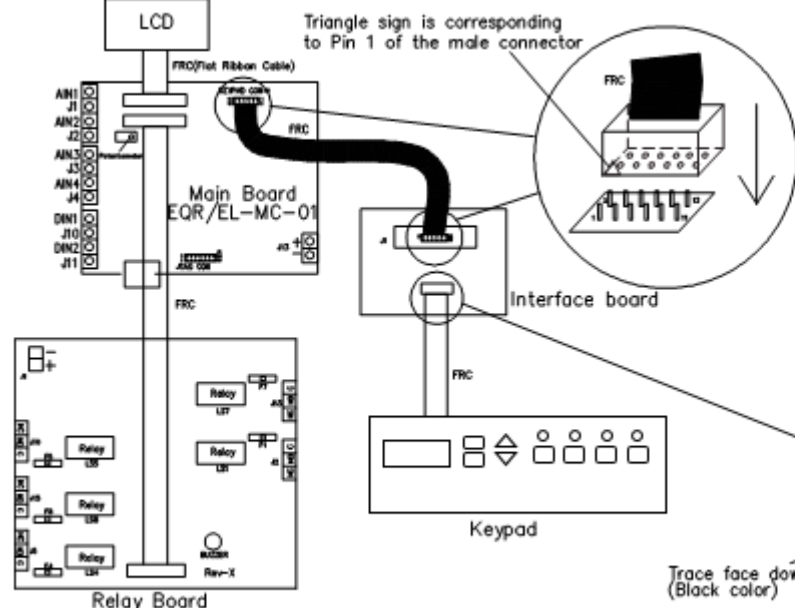
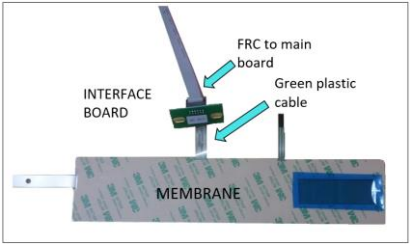
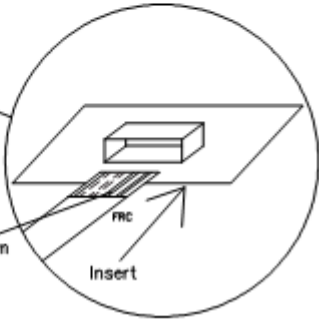
Figure 1-5

Problem 2: Blank LCD

Cause	Corrective Action
Connection problem	• Check LCD large Flat Ribbon Cable connecting the main board with the display at the main board end. • Check if the cable has been inserted properly into its socket on the main board. This is a Keyed connector with locking arms. The arms should be pointed in the same direction as the cable. See Figure 2-1 below. • If the LCD Display is not on, check power to main board as detailed in Problem 1.  Figure 2-1
Contrast problem	• Adjust the potentiometer on main board by rotating the top metal part using flat screw driver to achieve the best LCD contrast. Counter clock-wise direction will increase the contrast. See Figure 2-2. • If the LCD remains blank, replace it.  Figure 2-2
Defective LCD	• Connect a new LCD to the LCD port on main board. • If the new LCD functions properly, this indicates the old one is Defective. • If the new LCD does not working, replace the Large Flat Ribbon Cable. • If the LCD still does not work replace the main board.

Problem 3: Inoperative buttons

Cause	Corrective Action
Connection problem	• Does the LCD display work? If not go to Problem 1 and make sure the main board has power. • Ensure that the small Flat Ribbon Cable is connected to the main board. The triangle sign on the female connector of the cable indicates PIN number 1. • Interface board and membrane/keypad are located behind the blue panel, underneath the metal light deflector. • Turn off the cabinet remove the bulbs and the clips that hold the bulbs (bulb connections pull off). • Check connections between the cable from the main board and the Interface board. • Check the green plastic cable from the Interface board to the Display board.

Cause	Corrective Action
	
	 
Defective Cable and / or Interface Board and / or keypad	• Replace them one by one, to check which one(s) among them is/are Defective. • Replace the Defective part(s).

Problem 4: Blower doesn't function

Cause	Corrective Action
Motor Blower Fan is not running	• Switch on the Fan by pressing Fan button on control panel. • Enter the Fan PIN number if required (default is 0001) • The LED for FAN should illuminate and the FAN should start. • If the FAN does not operate, proceed to next step. • If the LED on the cabinet front does not illuminate refer to Problem 1.
DC Voltage Connection problem to relay board	• Measure the incoming voltage on the Relay Board at terminal J 1 (Note polarity, BLUE cable closest to edge is negative -). • Voltage should be between 10.8 – 13.2VDC. • If voltage is out of range or no voltage, check connection between SMPS (power supply) and relay board. • If voltage is correct, proceed to next step.

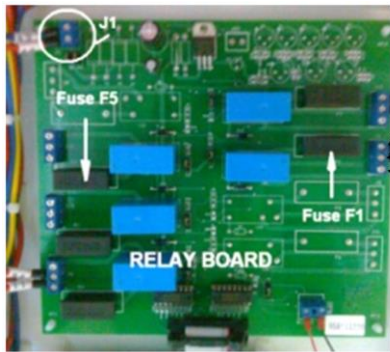
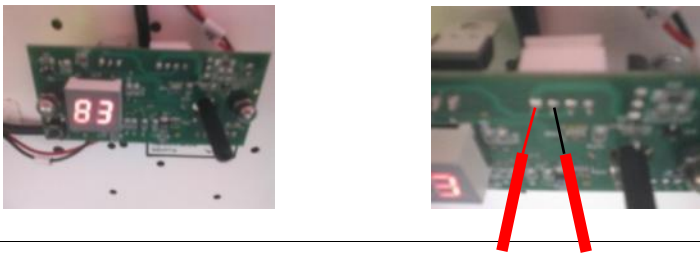
Fuse F1 is blown or circuit is not energized	• Check AC voltage input to the Relay board. • Ensure the Motor Blower FAN button and LED is energized. • Locate LS1 circuit and measure AC voltage between large BLUE (Neutral) terminal block and the end terminal (with Red cable) on the three terminal strips (J2 on relay board). See Figure 4-1. • If there is voltage to the end terminal (Terminal with cable) check the central terminal (Normally Open Circuit) to the BLUE (Neutral) terminal block. • When there is no voltage at the central terminal, it indicates a blown fuse or the circuit is not energized. • Check Fuse F1 on relay board. See Figure 4-1 below. • If fuse F1 is blown, as temporary solution, use F5 (spare) to replace F1. • If fuse F1 is okay, check the voltage between the BLUE (Neutral) terminal block and the end terminal on the LS1 circuit (J2 on relay board) • Voltage at the end terminal with no cable indicates the F1 fuse is good and the circuit is not energized.  Figure 4-1
Defective ECMS motor speed controller	• Check Voltage supply to the ECMS, check from DC supply that one pin must be 12 volt. • If there is no voltage supplied to the ECMS, check the cable connection. • If cable connection is fine, check voltage output of the DC power supply. • If there is no DC Power supply output, the DC power supply is faulty and needs replacement. • If there is voltage supplied to ECMS, proceed to the next step. • Check the display of ECMS whether or not there is any reading. • If ECMS displays any value, try to turn the potentiometer to maximum and minimum to check for response. • If ECMS responds to the adjustment, check PWM output with the multimeter. • If ECMS produces PWM output, ECMS is likely not defective. • If ECMS does not produce PWM output, ECMS is defective and needs replacement. • If ECMS does not respond after adjustment attempt with the potentiometer, ECMS is defective and needs replacement. • If ECMS does not display any value, ECMS is defective and needs replacement. • When the standby mode is turned on, the ECM blower should work at a lower speed. If the ECM blower increases in speed (measured or through displayed value) or drops lower than low than the expected speed, the ECMS is defective and need to be replaced. 

Figure 4-2. ECMS



Figure 4-4. ECMS no Display

Figure 4-3. Measure ECMS Supply

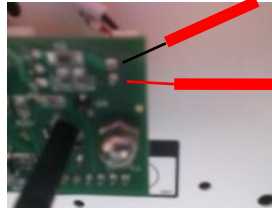


Figure 4-5. Measure PWM Output

Motor
failure

Warning: The cabinet *MUST* be decontaminated before opening the Panel to access the FAN.

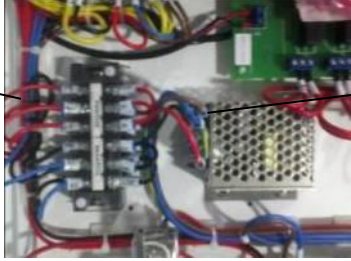
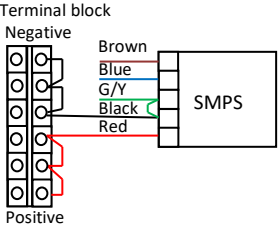
- Check for physical damage.
- Will the motor rotate?
- Check wiring.
- Replace the blower.

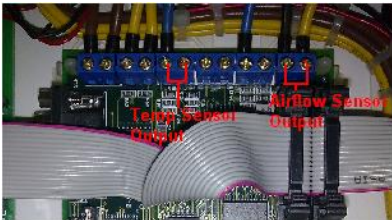
Problem 5: Airflow Failure (AIRFAIL) – alarm is triggered

Cause	Corrective Action
External air interference	• The airflow sensor is located at the top of the cabinet over the exhaust filter • Ensure that there are no external sources of airflow disturbance close to the sensor (vents, lights, ceiling too close, etc.) • If there is no external air interference, proceed to next step.
Blocked cabinet grille	• Ensure that the air grilles inside the cabinet are not blocked and the cabinet is not excessively loaded. • If there is no blockage of the air grilles, proceed to next step.
Air Flow Sensor loose or not positioned correctly	• Locate the airflow sensor at the top of the cabinet 1. Ensure that the sensor area is not blocked and its through-holes are perpendicular to the filter's surface. 2. Exhaust sensor is attached tightly to the sensor housing. 3. There is no gap between sensor housing and filter surface. 4. Sensor housing is installed tightly to the cabinet.
Low building supply voltage (if new cabinet, check this first)	• Open electrical box and find the display on the ECMS in Figures 5-1 • NOTE: A qualified Certifier should make all ECMS adjustments. • For a new cabinet, if the motor is lower than the nominal value recorded in the factory test report, adjust the ECMS to get output to blower as specified in the Test Report. • For cabinet that has been used for more than 1 year, refer to last ECMS Nominal Display from last certification report. Should this not be available, measure the actual airflow on cabinet and increase the EMCS percentage display accordingly to achieve nominal airflow velocity. • If the supply voltage to the ECMS remains at 12 VDC, proceed to next step.



Figure 5-1

Defective airflow sensor power supply –Switch Mode Power supply(SMPS)	• Cabinet must be turned on to perform this test. • Locate the terminal block. • On the side of the terminal block measure the DC voltage between red and Black Measure on the side away from the SMPS See Figure 5-2 and 5-3. • The voltage should be in range of +12VDC $\pm 10\%$. • You have to measure voltage on the other side of terminal block and SMPS. • If out of range, check incoming power to the SMPS – check terminal where the cord is connected. The input of SMPS should be $\pm 10\%$ of the cabinet incoming voltage • If you have input voltage but no output voltage replace the SMPS.
SMPS output Red – Black/blue is 12VDC	 Figure 5-2 SMPS input Brown-Blue-Green Yellow is 115V AC or 230V AC  Figure 5-3
Cabinet requires the air flows to be set	• Refer to the Test Report for this model and set the air flows to the required air flows values. This information is also contained on the label at the bottom of the BLUE front panel on the Left side as you face the cabinet.
Calibration	• Re-calibrate the microprocessor. Follow the calibration procedure in the test report. Warning: <i>To be carried out only by an authorized person.</i> CHECKING CALIBRATION DATA • After calibration is done, go to main display by pressing MENU button twice. • Press SET/Diagnostic button, LCD will display the mode and software version that is currently being used on the cabinet. Example: DIAGNOSTIC MODE: NORMAL CP107 VERSION 1.6 • Press DOWN button until you find the following: ○ TEMPERATURE: Shows actual ambient temperature ○ ADC IFF: ADC value for Fail Point Inflow – calculated using offset based on Inflow Nominal Point. ○ ADC IFN: ADC value for Nominal Point Inflow – set during sensor calibration. ○ ADC IFA: ADC value for Actual Inflow – showing real time sensor reading. ○ ADC IF0: ADC value for factory calibrated Zero Point Inflow (no inflow)-set during zero calibration. ○ ADC IF1: ADC value for factory calibrated Fail Point Inflow. ○ ADC IF2: ADC value for factory calibrated Nominal Point Inflow. ○ DFN: Nominal value of Down flow velocity - keyed in during field calibration. ○ CONSTANT: value for Airflow sensor constant - keyed in during sensor calibration. ○ CALIB TEMP: Temperature when the factory calibration was performed. ○ ADC TEMP: ADC value for TEMPERATURE. Note: <i>Depending on preset unit at SETTINGS>MEASUREMENT UNIT> (Metric or Imperial), the units for temperature and velocity will change accordingly. For Metric-°C, m/s while for Imperial-°F, fpm. Factory setting is at Metric.</i> • In order to prevent an AIRFAIL alarm or unstable velocity displayed on LCD ensure the followings after the calibration done: 1. CONSTANT value is correct. Check it with constant written on sensor body or inside

	Replaceable Component Record List which come with test report. - TEMPERATURE shows room temperature. If not, refer to next possible cause, part B. - ADC IF0 < ADC IF1 < ADC IF2. If not please re-do calibration in correct sequence: SET CONSTANT → SET ZERO → CALIB. If CALIBRATION ERROR is encountered, please refer to refer to next possible cause, part A. - ADC IF0 < ADC IFN. If not please re-do calibration in correct sequence: SET CONSTANT → SET ZERO → CALIB. If CALIBRATION ERROR is encountered, please refer to next possible cause, part A. - Just after calibration, ADC IFA value should be close to ADC IFN value. If not, please adjust the speed controller to make both values closer by at least ± 20 units.
Sensor Failure	A. Air Flow Sensor failure / misalignment - Locate the main board. - Using DVM, check the sensor's output voltage (DC) at channel AIN1(+) and J1(-) on main board (See Figure 5-4 below). - The voltage should increase if motor voltage is increased and decrease if motor voltage is decreased. - If your observation is different, check sensor connection on the cabinet and at connector D pin 9,10,11,12 on electrical panel for any loose connection. - If connection is okay but sensor is still not working properly, replace it. - If airflow sensor is working properly, proceed to next step. B. Temperature Sensor Failure - Temperature sensor is located close to airflow sensor See figure 5-6. - Using DVM, check the temperature sensor's output voltage (DC) at channel AIN4(-) and J4(+) on main board (See figure 5-4 below). 0.01 VDC should represent 1°C. For example, if output is 0.25 VDC, it indicates temperature of 25 °C. Tolerance of ± 1 °C is acceptable. - Replace temperature sensor if the reading is wrong. - If temperature sensor is working properly, proceed to next step.
  	
Figure 5-4 Figure 5-5 Figure 5-6	

Problem 6: Excessive Blower Noise

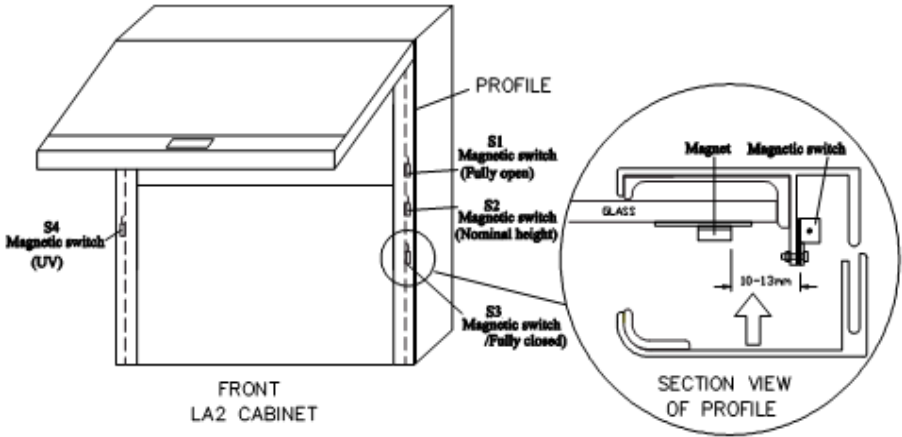
Cause	Corrective Action
Resonance	NOTE: Requires a Certifier - Locate the motor speed controller. - Measure the motor voltage and record this number. - Increase the motor speed by 5 – 10 VAC. Does the noise change or go away? - If the noise is gone, measure the actual airflow velocity. Requires a Certifier - If the velocity is still in the acceptable range, maintain it. - If velocity is out of acceptable range, or if noise persists, go to next step.
Loose motor or impeller wheel mounting	- Refer to Problem 2 - Faulty Capacitor section, to check the capacitor, and replace if necessary. Note: Decontaminate cabinet first before accessing the blower - Open the blower access panel and check if the blower mounting bolts are tight - Does the motor rotate freely without noise? - If motor is physically damaged, replace it.

Problem 7: Light always OFF

Cause	Corrective Action
Sash in SASH ALARM state	- Is the Light LED on the front panel ON? - If not press the Light button and make sure the sash is in the Normal operating position (Ready Position Normal Operating height) Note: <i>The sash must be at the Normal Operating position.</i> - Is the cabinet in Warm Up? Note: <i>The fluorescent light will not light if the cabinet is in Warm Up unless the cabinet is in Quick Start Mode.</i> - Move the sash to Ready position (normal operating height). - Switch on the light by pressing LIGHT button on the membrane. - If the LED is ON the keypad and main board are functioning correctly. - If fluorescent light does not light with the window in the proper position proceed to next step. - If the LIGHT LED still does not light, skip to Ready magnetic switch issues.
Burned out fluorescent tube	- Replace the faulty fluorescent tube. - Fluorescent tubes are located inside the blue panel. - The white lamp holder can be pulled off from the tubes.
Defective fluorescent ballast or relay board issues	- Check voltage coming to the Relay board. - Locate LS7 circuit and measure AC voltage between BLUE (NEUTRAL) terminal block and the end terminal (with Red cable) on the three terminal strip (J13 on relay board). - If there is voltage to the end terminal (Terminal with cable) check the central terminal (Normally Open Circuit) to the BLUE (NEUTRAL) terminal block. - No voltage at the central terminal indicates a blown fuse or the circuit is not energized. - Check Fuse F7 on relay board. See Figure 7-1 below. - If fuse F7 is blown, as temporary solution, use F5 (spare) to replace F7. - If fuse F7 is okay, check the voltage between the BLUE (NEUTRAL) terminal block and the end terminal on the LS7 circuit (J13 on relay board). - Voltage at the end terminal with no cable indicates the F7 fuse is good and the circuit is not energized. - Make sure the LIGHT LED is on. - If the ballast has power and the bulbs are new the problem is either connections or the ballast. - Check Connector C connections. If there are no loose terminals replace the ballast.



Figure 7-1

Ready magnetic switch issues	• With the LIGHT LED ON and the sash at the Normal operating position (Ready). • Does the Display indicate SASH: OK or SASH: NO! • SASH: NO indicates the sash is not at the proper position and the LIGHT LED should not be on. This indicates a main board problem or a magnetic switch detection problem. • To access the magnetic switches for the sash positions remove the Right sash window profile cover as you face the cabinet. • There are 3 magnetic switches inside the right profile. See Figure 7-2 below. The one in the middle is for Ready position. • There is a magnet attached to the glass beside the middle switch. • Shift the switch or magnet position so the distance between them is between 10-13 mm (3/8 to 1/2") as explained in Figure 7-2 below. If the distance is too far, the switch may not be able to detect the magnet. • If the LCD still displays "SASH: NO!", the switch may be defective or there is a wiring connection problem. To check the connections proceed to the next step.  Figure 7-2
Connection Problem	• Locate Connector C and D and check male and female side for any bad connection. • If the fluorescent lamp still does not light with the LIGHT LED ON return to start of Problem 7 and repeat the troubleshooting process.

Problem 8: UV Light always OFF

Cause	Corrective Action
Sash not in UV state	• Lower the sash to UV mode position. • If SASH FULLY CLOSED is not displayed on the LCD, please refer to the Magnetic Switch troubleshooting section. • Switch on the UV lamp by pressing the UV button. • If SASH FULLY CLOSED is displayed on LCD but UV is not ON after pressing UV button, refer to the Magnetic Switch troubleshooting section.
Faulty UV tube	• Replace the faulty UV tube.

Faulty UV ballast	• Check voltage coming to the Relay board.
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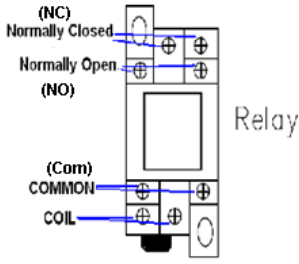
	• Locate LS8 (J15 Terminal) circuit and measure AC voltage between large BLUE (Neutral) terminal block and the end terminal (with Red cable) on the three terminal strip (J15 on relay board). • If there is NO Power to the end terminal (Terminal with cable) refer to the next step NO power to UV Relay circuit LS8. • If there IS voltage to the end terminal (Terminal with cable) check the central terminal (Normally Open Circuit) to the BLUE (Neutral) terminal block. • No voltage at the central terminal indicates a blown fuse or the circuit is not energized. • Check Fuse F8 on relay board. See Figure 8-1 below. • If fuse F8 is blown, as temporary solution, use F5 (spare) to replace F8. • If fuse F8 is okay, check the voltage between the BLUE(NEUTRAL) terminal block and the end terminal on the LS8 circuit (on J15 terminal of relay board) • Voltage at the end terminal with no cable indicates the F8 fuse is good and the circuit is not energized • Make sure the UV LED is on. • If the ballast has power and the UV lamp is new, the problem is either cable connections or the UV ballast. • Check Connector E (115V) or B (230V) cable connections. If there are no loose terminals replace the UV ballast.
No Power to UV Relay circuit LS8 and K2 relay	• No power to LS8 indicates K2 Relay is open. • Check magnetic UV interlock switch on Left side (as you face the cabinet) of sash. • Check the AC voltage to and from the K2 relay • No voltage indicates a wiring issue between the RED (Live) power in and the K2 relay. • If voltage to the K2 relay is present, but no voltage out of the K2 relay, this indicates the relay is not energized or defective. • Check the DC voltage to the coil of the K2 relay. Voltage should be 12VDC $\pm$ 10% • No voltage indicates the magnetic switch on the Left side is not closed. Check the wiring to the switch Connector D-6 and D-3. • To simulate a closed switch jumper between connector D-6 and D-3 at Connector D • If the relay does not energize check the wiring from the K2 coil to Connector D and the neutral wiring from K2 to the DC Power supply. • If wiring connections are OK, visually inspect K2 relay for any burnt or flash marks. • If burnt or flash marks are present, replace the relay. • If no burnt or flash marks are found, turn off the cabinet. • Disconnect relay cables and check for the continuity at K2 relay terminals. See figure 8-2. NC to Com has continuity while, NO to Com doesn't have continuity. Replace the relay if any of the terminals are found with wrong continuity configuration.  

Figure 8-1

Figure 8-2

UV magnetic switch is defective or misaligned	• Move sash to Fully Closed position. The LCD should show SASH: FULLY CLOSED. • If the LCD display shows "SASH: NO!" the sash is not fully closed. • Check the magnetic switches. There are 3 magnetic switches inside the right profile. See Figure 7-2 above. The bottom one is for UV Mode position. • There is a magnet attached to the glass beside the bottom switch. • Shift the switch or magnet position so the distance between them is between 10-13 mm (3/8" to 1/2") as explained in Figure 7-2 above. If the distance is too far, the switch may not be able to detect the magnet. • If the LCD still showing "SASH: NO" check the connections at Connector D pins D-4 and D-5. • To simulate a closed U/V Switch jumper between Connector D pins D-5 and D-4. • If this does not turn on the UV go back to the start of this procedure and check it again.
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Problem 9: Electrical socket always OFF

Cause	Corrective Action
Connection Problem	• Is the OUTLET LED Illuminated? If not press the Outlet button. • On 115 VAC cabinets, check the GFCI outlets. If the LED on the GFCI is on, reset the GFCI by pressing on the center reset button. A GFCI out with the LED on the outlet itself indicates the outlet has power, but is tripped. • Check voltage coming to the Relay board. • Locate LS4 (J6 Terminal) circuit and measure AC voltage between Large BLUE (Neutral) terminal block and the end terminal (with Red cable) on the three terminal strip (J6). • If there is voltage to the end terminal (Terminal with cable) check the central terminal (Normally Open Circuit) to the BLUE (Neutral) terminal block. • No voltage at the central terminal indicates a blown fuse or the circuit is not energized. • Check Fuse F4 on relay board. See Figure 9-1 below. • If fuse F4 is blown, as temporary solution, use F5 (spare) to replace F4. • If fuse F4 is OK, check the voltage between the BLUE (NEUTRAL) terminal block and the end terminal on the LS4 circuit (J6 on relay board). • Voltage at the end terminal with no cable indicates the F4 fuse is good and the circuit is not energized. Make sure the OUTLET LED is on. • If there is power to the relay and it is energized Check Connector E (115V) or B (230V) pin 1 and 4 to the Connector E (115V) or B (230V) pin 2 and 5. • If there is power through Connector E, remove outlets from wall. • Check incoming power to the plug in the cable to the outlet. • No power indicates a wiring issue. Go back and check voltage from Connector E (115V) or B (230V). • If there is no voltage to the relay check the Auxiliary line power cord (if provided with 2 power cords) and make sure it is connected to an outlet with the correct voltage. • If the outlet and cord are connected check Connector C pin C-3 Live and C-6 Neutral for incoming power. Use the RED and BLUE terminal strip to confirm there is power and a good neutral. • No power indicates a problem with the wiring between the cord and the connector. • Check for loose or faulty connection between the power cord, IEC inlet and Connector C.

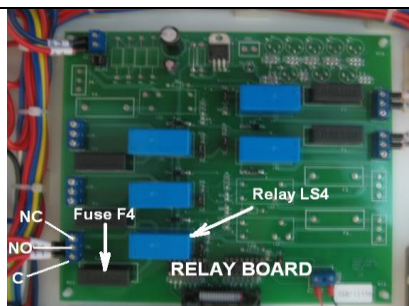
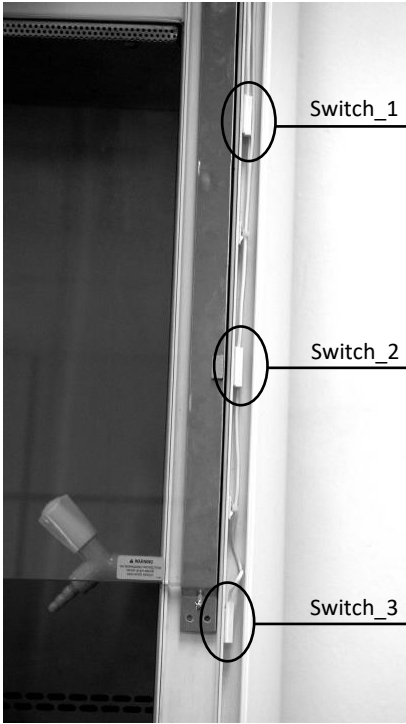
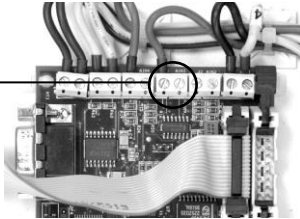


Figure 9-1

Problem 10: Contaminated Sample/Lab

Cause	Corrective Action
Blocked air grille/ Cabinet overloaded with too many items.	Note: Contaminated samples are possibly caused by: - Poor microbiological technique. - Samples contaminated elsewhere prior to being worked in the BSC. Items to check on the BSC: • Make sure that the front and back air grilles are not blocked. • Remove all equipment and other items from the cabinet and wipe down the cabinet including the area under the work surface. • Wipe down everything that is returned to the BSC. • If the contamination problem continues have the cabinet certified including air flows and filter leak tests.

3.2. Software Troubleshooting

Error Code	Corrective Action
Sash: Error Position	• Magnetic switches are located inside right profile cover. See drawing B-1 below.
    B-1	

Error Code

Corrective Action

The diagram illustrates the front view of a sash window. A central rectangle represents the window frame, labeled "Sash Window (front view)". At the top-left corner of the frame is a small black rectangle labeled "Magnet". At the top-right corner is another small black rectangle labeled "Magnet". Along the right edge of the frame, there are three small squares labeled "Mag. Switch 1", "Mag. Switch 2", and "Mag. Switch 3" from top to bottom. Along the left edge, there is one small square labeled "Mag. Switch 4". Below the left side of the window frame, a vertical double-headed arrow indicates the "Sash movement range".

B-2

Magnetic Switch 1	Magnetic Switch 2	Magnetic Switch 3	Magnetic Switch 4	Description
Close	Open	Open	Open	Sash is fully opened, ALARM is given. ALARM can be muted.
Open	Open	Close	Close	Cabinet sash is at nominal height, ready to use.
Open	Close	Open	Open	Sash is fully closed. UV can be opened.
Open	Open	Open	Open	SASH is at unsafe state, ALARM is given. ALARM cannot be muted.
Rest Conditions				SASH:ERROR POSITION detected

Table 2

SENSOR
UNCALIBRATED

- Calibrate the controller. Refer to test report to calibrate.

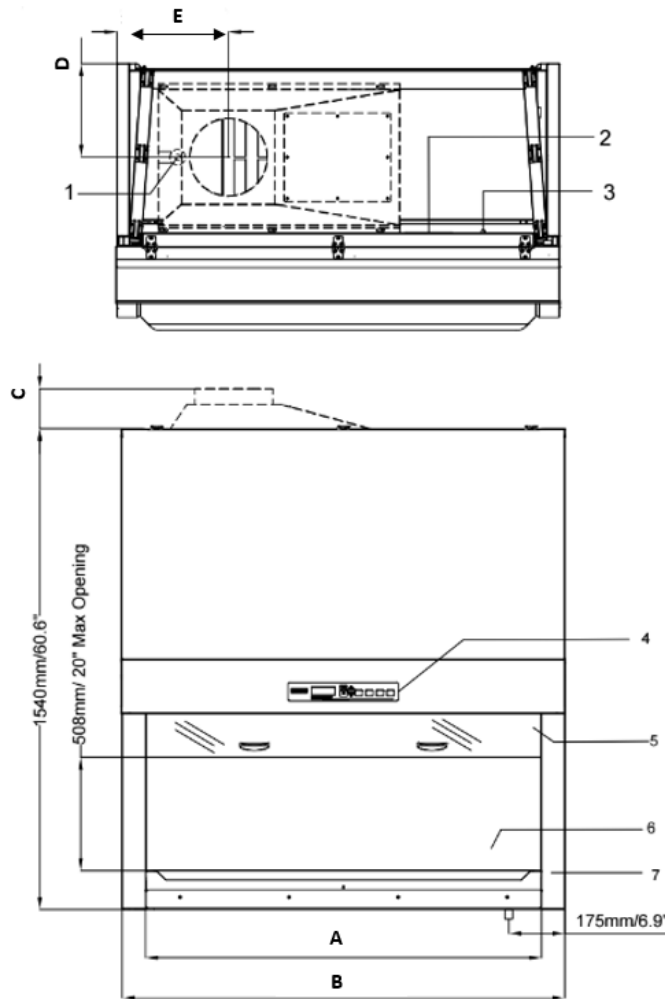
3.3. Sash Operation Mode Summary

Sash Position	Function Available	Normal Or Quickstart Mode and Airfail Alarm On	Normal Or Quickstart Mode and Airfail Alarm Off	Maintenance Mode
READY	Sash Position Detection	Yes	Yes	Yes
	Fan Control	Yes, with WARM UP & PURGING time, FAN PIN required	Yes, with WARM UP & PURGING time, FAN PIN required	Yes, without WARM UP & PURGING time, no PIN required
	Light Control	Yes	Yes	Yes
	Socket Control	Yes	Yes	Yes
	UV Control	Interlocked	Interlocked	Interlocked
	MENU Access	Yes, USER or ADMIN PIN required	Yes, USER or ADMIN PIN required	Yes, no PIN required
	Pressing Set button	Complete Diagnostic Message	Complete Diagnostic Message	Complete Diagnostic Message
	Timer	Yes	Yes	No
	Current Time Display	Yes	Yes	Yes
	Air Velocity Display	Yes	Yes	Yes
	If not calibrated, displaying "Sensor Uncalibrated"			
SASH ALARM	Air Velocity Display Status	Yes	Yes	No
	Air Fail checking	Yes, Alarm and display alert if air fails	Yes, Alarm and display alert if air fails	No
	Sash Position Detection	Yes	Yes	Yes
	Current Time Display	Yes	Yes	Yes
	Sash Alarm	Yes, not mutable	Yes, not mutable	Not Applicable
	Fan Control	Yes, without WARM UP & PURGING time, no PIN required	Yes, without WARM UP & PURGING time, no PIN required	Yes, without WARM UP & PURGING time, no PIN required
	Light Control	No	No	Yes
	Socket Control	Yes	Yes	Yes

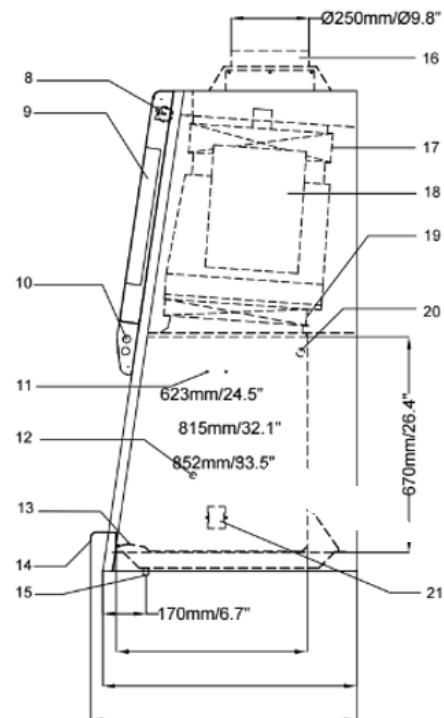
	UV Control	Interlocked	Interlocked	Interlocked
	Air Velocity Display	Yes	Yes	Yes
	Air Velocity Display Status	Yes	Yes	No
	Air Fail checking	Yes, display alert if air fails	Yes, display alert if air fails	No
FULLY OPEN	Sash Position Detection	Yes	Yes	Yes
	Current Time Display	Yes	Yes	Yes
	Alarm	Yes, mutable for 5 minutes	Yes, mutable for 5 minutes	Not Applicable
	Fan Control	Yes, with WARM UP & PURGING time, FAN PIN required	Yes, with WARM UP & PURGING time, FAN PIN required	Not Applicable
	Light Control	Yes	Yes	Yes
	Socket Control	Yes	Yes	Yes
	UV CONTROL	Interlocked	Interlocked	Interlocked
FULLY CLOSED	Sash Position Detection	Yes	Yes	Yes
	SWITCH OFF Fan requested	Yes	Yes	No
	Fan Control	Interlocked	Interlocked	Yes
	Light Control	Interlocked	Interlocked	Yes
	Socket Control	Yes	Yes	Yes
	UV Control	Yes	Yes	Interlocked

Chapter 4 - Engineering Details

4.1. Engineering Drawing



	A	B	C	D	E
3'	970mm 38.2"	1115mm 43.9"	128mm 5.0"	288mm 11.3"	370mm 15.0"
4'	1270mm 50.0"	1420mm 55.9"	133mm 5.2"	288mm 11.3"	530mm 20.9"
5'	1570mm 61.8"	1725mm 67.9"	137mm 5.4"	288mm 11.3"	610mm 24.0"
6'	1870mm 73.6"	2030mm 79.9"	138mm 5.4"	288mm 11.3"	711mm 28.0"
8'	2440mm 96.0"	2600mm 102.4"	140mm 5.5"	288mm 11.3"	1300mm 51.2"



- | | | |
|--|---------------------------------------|--|
| 1. Sensor Box | 8. Tubular Motor (If Any) | 16. Exhaust Collar (Optional) |
| 2. RS232 Port and Zero Voltage Relay Contact | 9. Electrical Panel | 17. Exhaust ULPA Filter |
| 3. Power Inlet | 10. Fluorescent Lamp(s) | 18. Blower |
| 4. Sentinel Gold Controller | 11. IV Bar Retrofit Kit Provision | 19. Downflow ULPA Filter |
| 5. Sash Window | 12. Plugged Service Fixture Provision | 20. UV Lamp Retrofit Kit Provision |
| 6. Back Wall | 13. Worktray | 21. Electrical Outlet Retrofit Kit Provision |
| 7. Removable Side Dress Panel | 14. Armrest | |
| | 15. Drain Pan | |

4.2. General Specifications

	LA2-3A_-E	LA2-4A_-E	LA2-5A_-E	LA2-6A_-E	LA2-8A_-E
External Dimensions (W x D x H)	1115x852x1540mm 44.0" x 33.5" x 60.6"	1420x852x1540mm 56.0" x 33.5" x 60.6"	1725x852x1540mm 68.0" x 33.5" x 60.6"	2030x852x1540mm 80.0" x 33.5" x 60.6"	2600x852x1540mm 102.4" x 33.5" x 60.6"
Gross Internal Dimensions* (W x D x H)	970x623x670mm 38.2" x 24.5" x 26.4"	1270x623x670mm 50.0" x 24.5" x 26.4"	1570x623x670mm 61.8" x 24.5" x 26.4"	1870x623x670mm 73.6" x 24.5" x 26.4"	2440x623x670mm 96.0" x 24.5" x 26.4"
Usable Work Area	0.45 m ² (4.8 sq.ft.)	0.60 m ² (6.5 sq.ft.)	0.75 m ² (8.1 sq.ft.)	0.90 m ² (9.7 sq.ft.)	1.2m ² (13 sq.ft.)
Tested Opening	229 mm (9")	229 mm (9")	229 mm (9")	203 mm (8")	203 mm (8")
Working Opening	274 mm (10.8")	274 mm (10.8")	274 mm (10.8")	248 mm (9.8")	248 mm (9.8")
Average Airflow Velocity	Inflow	0.53 m/s (105 fpm)			
	Downflow	0.35 m/s (70 fpm)	0.35 m/s (70 fpm)	0.35 m/s (70 fpm)	0.33 m/s (65 fpm)
Airflow Volume	Inflow	424 m ³ /h (251 cfm)	555 m ³ /h (328 cfm)	686 m ³ /h (406 cfm)	724 m ³ /h (426 cfm)
	Downflow	628 m ³ /h (363 cfm)	822 m ³ /h (476 cfm)	1016 m ³ /h (588 cfm)	1210 m ³ /h (700 cfm)
	Exhaust	424 m ³ /h (251 cfm)	555 m ³ /h (328 cfm)	686 m ³ /h (406 cfm)	724 m ³ /h (426 cfm)
	Required Exhaust With Optional Thimble Exhaust Collar	529 m ³ /h (311 cfm)	764 m ³ /h (450 cfm)	1116 m ³ /h (657 cfm)	1164 m ³ /h (685 cfm)
	Static Pressure For Optional Thimble Exhaust Collar	32 Pa / 0.12 in H ₂ O	49 Pa / 0.19 in H ₂ O	62 Pa / 0.24 in H ₂ O	79 Pa / 0.31 in H ₂ O
ULPA Filter Typical Efficiency		≥ 99.999% at particle size between 0.1 to 0.3 microns per IEST-RP-CC001.3 / H14 per EN 1822			
Cabinet Construction	Main Body	Electro-galvanized steel with white oven-baked epoxy-polyester Isocide™ antimicrobial powder-coated finish, 1.5 mm (0.06") / 16 gauge thick			
	Work Zone	Stainless steel Type 304 with No.4 finish, 1.5 mm (0.06") / 16 gauge thick			
Sound Emission**	NSF / ANSI 49	62.5 dBA	63 dBA	63.5 dBA	64 dBA
	EN 12469	59.5 dBA	60 dBA	60.5 dBA	61 dBA
Fluorescent Lamp Intensity		>1230 Lux (>114 foot-candles)	>1400 Lux (>130 foot-candles)	>1070 Lux (>100 foot-candles)	>1230 Lux (>114 foot-candles)
Electrical	Full Load Amps 230V	4.5 A	5.5 A	5.7 A	6 A
	Full Load Amps 115V	9 A	11 A	11.5 A	12 A
	Heat Load	853 BTU/Hr.	972 BTU/Hr.	1177 BTU/Hr.	1297 BTU/Hr.
Net Weight ****		243 kg / 536 lbs.	283 kg / 624 lbs.	350 kg / 772 lbs.	426 kg / 939 lbs.
Shipping Weight ****		292 kg / 644 lbs.	345 kg / 761 lbs.	410 kg / 904 lbs.	486 kg / 1072 lbs.
Shipping Dimensions, Maximum (W x D x H) ****		1200x950x1900mm 47.2" x 37.4" x 74.8"	1550x950x1900 mm 61.0" x 37.4" x 74.8"	1950x950x1900mm 76.8" x 37.4" x 74.8"	2150x950x1900mm 84.6" x 37.4" x 74.8"
Shipping Volume, Maximum ****		2.17 m ³ (77 cu.ft.)	2.80 m ³ (99 cu.ft.)	3.52 m ³ (124 cu.ft.)	3.88 m ³ (137 cu.ft.)

* Depth includes the remove-able arm rest and front cover. When they are removed, depth is 790 mm (31.1").

** Noise reading in open field condition / anechoic chamber.

*** Additional voltages may be available; contact Esco for ordering information.

**** Cabinet only, excludes optional stand.

		LA2-3L_-E	LA2-4L_-E	LA2-5L_-E	LA2-6L_-E
External Dimensions (W x D x H)		1115x852x1540mm 44.0" x 33.5" x 60.6"	1420x852x1540mm 56.0" x 33.5" x 60.6"	1725x852x1540mm 68.0" x 33.5" x 60.6"	2030x852x1540mm 80.0" x 33.5" x 60.6"
Gross Internal Dimensions* (W x D x H)		970x623x670mm 38.2" x 24.5" x 26.4"	1270x623x670mm 50.0" x 24.5" x 26.4"	1570x623x670mm 61.8" x 24.5" x 26.4"	1870x623x670mm 73.6" x 24.5" x 26.4"
Usable Work Area		0.45 m ² (4.8 sq.ft.)	0.60 m ² (6.5 sq.ft.)	0.75 m ² (8.1 sq.ft.)	0.90 m ² (9.7 sq.ft.)
Tested Opening		229 mm (9")	229 mm (9")	229 mm (9")	203 mm (8")
Working Opening		274 mm (10.8")	274 mm (10.8")	274 mm (10.8")	248 mm (9.8")
Average Airflow Velocity	Inflow	0.45 m/s (88 fpm)			
	Downflow	0.30 m/s (59 fpm)			
Airflow Volume	Inflow	272 m ³ /h (160 cfm)	356 m ³ /h (209 cfm)	440 m ³ /h (259 cfm)	524 m ³ /h (308 cfm)
	Downflow	653 m ³ /h (384 cfm)	855 m ³ /h (503 cfm)	1056 m ³ /h (621 cfm)	1258 m ³ /h (740 cfm)
	Exhaust	272 m ³ /h (160 cfm)	356 m ³ /h (209 cfm)	440 m ³ /h (259 cfm)	524 m ³ /h (308 cfm)
ULPA Filter Typical Efficiency		≥ 99.999% at particle size between 0.1 to 0.3 microns per IEST-RP-CC001.3 / H14 per EN 1822			
Cabinet Construction		1.5 mm (0.06") / 16 gauge Electrogalvanized steel with Isocide oven-baked epoxy-polyester powder coating			
Sound Emission**	NSF / ANSI 49	62.5 dBA	63 dBA	63.5 dBA	64 dBA
	EN 12469	59.5 dBA	60 dBA	60.5 dBA	61 dBA
Fluorescent Lamp Intensity		>800 Lux (>75 foot-candles)	>800 Lux (>75 foot-candles)	>800 Lux (>75 foot-candles)	>800 Lux (>75 foot-candles)
Electrical	220 – 240 V AC, 50 Hz, 1Φ	LA2-3L1-E LA2-3L1-E-M	LA2-4L1-E LA2-4L1-E-M	LA2-5L1-E LA2-5L1-E-M	LA2-6L1-E LA2-6L1-E-M
	110 – 125 V AC, 60 Hz, 1Φ	LA2-3L2-E LA2-3L2-E-M	LA2-4L2-E LA2-4L2-E-M	LA2-5L2-E LA2-5L2-E-M	LA2-6L2-E LA2-6L2-E-M
	220 – 240 V AC, 60Hz, 1Φ	LA2-3L3-E LA2-3L3-E-M	LA2-4L3-E LA2-4L3-E-M	LA2-5L3-E LA2-5L3-E-M	LA2-6L3-E LA2-6L3-E-M
Net Weight ****		243 kg / 536 lbs.	283 kg / 624 lbs.	350 kg / 772 lbs.	394 kg / 869 lbs.
Shipping Weight ****		292 kg / 644 lbs.	345 kg / 761 lbs.	418 kg / 921.5 lbs.	462 kg / 1019 lbs.
Shipping Dimensions, Maximum (W x D x H) ****		1200x950x1900mm 47.2" x 37.4" x 74.8"	1550x950x1900 mm 61.0" x 37.4" x 74.8"	1950x950x1900mm 76.8" x 37.4" x 74.8"	2150x950x1900mm 84.6" x 37.4" x 74.8"
Shipping Volume, Maximum ****		2.17 m ³ (77 cu.ft.)	2.80 m ³ (99 cu.ft.)	3.52 m ³ (124 cu.ft.)	3.88 m ³ (137 cu.ft.)

* Depth includes the remove-able arm rest and front cover. When they are removed, depth is 790 mm (31.1").

** Noise reading in open field condition / anechoic chamber.

*** Additional voltages may be available; contact Esco for ordering information.

**** Cabinet only, excludes optional stand.